

Exposé I. Presheaves

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In numbers 0 through 5 of this exposé we present the elementary, and most often well-known, properties of categories of presheaves.¹ These properties are used constantly throughout the rest of the seminar, and knowing them is essential for understanding the subsequent exposés. The proofs are immediate; for the most part they are omitted. In numbers 6 through 9 we take up a few themes that are used several times later on. The reader in a hurry may omit them on a first reading. Number 10 fixes the terminology used. Appendix 11 is due to N. Bourbaki.

1 Universes

A *universe* is a nonempty set² $\mathcal{U}$ satisfying the following properties:

- (U1) If $x \in \mathcal{U}$ and $y \in x$, then $y \in \mathcal{U}$.
- (U2) If $x, y \in \mathcal{U}$, then $\{x, y\} \in \mathcal{U}$.
- (U3) If $x \in \mathcal{U}$, then $\mathcal{P}(x) \in \mathcal{U}$.
- (U4) If $(x_i)_{i \in I}$ is a family of elements of $\mathcal{U}$ and $I \in \mathcal{U}$, then $\bigcup_{i \in I} x_i \in \mathcal{U}$.

From the preceding axioms one easily deduces the following properties:

- if $x \in \mathcal{U}$, then the singleton $\{x\}$ belongs to $\mathcal{U}$;
- if x is a subset of some $y \in \mathcal{U}$, then $x \in \mathcal{U}$;
- if $x, y \in \mathcal{U}$, the ordered pair $(x, y) = \{\{x, y\}, x\}$, in Kuratowski's definition, is an element of $\mathcal{U}$;
- if $x, y \in \mathcal{U}$, the union $x \cup y$ and the product $x \times y$ are elements of $\mathcal{U}$;
- if $(x_i)_{i \in I}$ is a family of elements of $\mathcal{U}$ indexed by an element $I \in \mathcal{U}$, then the product $\prod_{i \in I} x_i$ is an element of $\mathcal{U}$;
- if $x \in \mathcal{U}$, then $\text{card}(x) < \text{card}(\mathcal{U})$ strictly. In particular, the relation $\mathcal{U} \in \mathcal{U}$ is not satisfied.

Thus one may perform all the usual operations of set theory starting from elements of a universe without the final result ceasing to be an element of that universe.

The first use of the notion of universe is that it supplies a definition of the usual categories: the category of sets belonging to the universe $\mathcal{U}$ ($\mathcal{U}\text{-Set}$), the category of topological spaces belonging to $\mathcal{U}$, the category of commutative groups belonging to $\mathcal{U}$ ($\mathcal{U}\text{-Ab}$), the category of categories belonging to $\mathcal{U}$, and so on.

However, the only known universe is the set of symbols of the type $\{\{\emptyset\}, \{\{\emptyset\}, \emptyset\}\}$, etc. All the elements of this universe are finite sets, and this universe is countable. In particular, no

¹The reader may also consult SGA 3, I, §§1–3.

²This definition contradicts Bourbaki's appendix below. However, the interest of empty universes seems debatable.

universe is known that contains an element of infinite cardinality. We are therefore led to add to the axioms of set theory the axiom

(UA) *For every set x there exists a universe $\mathcal{U}$ such that $x \in \mathcal{U}$.*

Since the intersection of a family of universes is a universe, it follows immediately that every set is an element of a smallest universe. One can show that axiom (UA) is independent of the axioms of set theory.

We shall also add the axiom:

(UB) *Let $R\{x\}$ be a relation and let $\mathcal{U}$ be a universe. If there exists an element $y \in \mathcal{U}$ such that $R\{y\}$, then*

The consistency of axioms (UA) and (UB) relative to the other axioms of set theory is not proved and, it seems, not provable.

Let $\mathcal{U}$ be a universe and let $c(\mathcal{U})$ be the least upper bound of the cardinals of elements of $\mathcal{U}$; thus $c(\mathcal{U}) \leq \text{card}(\mathcal{U})$. The cardinal $c(\mathcal{U})$ has the following properties:

(FI) If $a < c(\mathcal{U})$, then $2^a < c(\mathcal{U})$.

(FII) If $(a_i)_{i \in I}$ is a family of cardinals strictly smaller than $c(\mathcal{U})$, and if $\text{card}(I)$ is strictly smaller than $c(\mathcal{U})$, then $\sum_{i \in I} a_i < c(\mathcal{U})$.

Cardinals satisfying properties (FI) and (FII) are called *strongly inaccessible cardinals*.

The cardinal 0 and the countably infinite cardinal are strongly inaccessible.

Axiom (UA) implies:

(UA') *Every cardinal is strictly bounded above by a strongly inaccessible cardinal.*

One can show (11) that, conversely, the consistency of (UA') implies the consistency of (UA), and that the consistency of these axioms implies the consistency of axiom (UB).

Call a set E *Artinian* if there is no infinite family $(x_n)_{n \in \mathbb{N}}$ such that $x_0 \in E$ and $x_{n+1} \in x_n$. One can then show [*loc. cit.*] that there is a one-to-one correspondence between strongly inaccessible cardinals and Artinian universes, defined as follows: to every strongly inaccessible cardinal c one associates the unique Artinian universe $\mathcal{U}_c$ such that

$$\text{card}(\mathcal{U}_c) = c.$$

If $c < c'$ are two strongly inaccessible cardinals, then

$$\mathcal{U}_c \in \mathcal{U}_{c'}.$$

In particular, the Artinian universes of cardinal smaller than a given cardinal form a well-ordered set for the membership relation. Axiom (UA') is equivalent to the axiom:

(UA'') *Every Artinian set is an element of an Artinian universe.*

We note that all the usual sets $(\mathbb{Z}, \mathbb{Q}, \dots)$ are Artinian sets.

2 $\mathcal{U}$ -categories. Presheaves of sets

1.0

In the rest of the seminar, unless the contrary is expressly stated, the universes considered will possess an element of infinite cardinality. Let $\mathcal{U}$ be a universe. A set is said to be $\mathcal{U}$ -small, or simply *small* when no confusion can result, if it is *isomorphic* to an element of $\mathcal{U}$. We also use the terminology small group, small ring, small category,³ etc. We shall often suppose, without explicit mention, that the schemes, topological spaces, indexing sets, etc. with which we work are in $\mathcal{U}$, or at least have cardinality in $\mathcal{U}$; however, many of the categories with which we shall work will not themselves be in $\mathcal{U}$.

Definition (1.1). Let $\mathcal{U}$ be a universe and let C be a category. We say that C is a $\mathcal{U}$ -category if, for every pair (x, y) of objects of C , the set $\text{Hom}_C(x, y)$ is $\mathcal{U}$ -small.

1.1.1

Let C and D be two categories, and let $\text{Fun}(C, D)$ denote the category of functors from C to D . The following assertions are immediate:

- (a) If C and D are elements of a universe $\mathcal{U}$ (respectively are $\mathcal{U}$ -small), then the category $\text{Fun}(C, D)$ is an element of $\mathcal{U}$ (respectively is $\mathcal{U}$ -small).
- (b) If C is $\mathcal{U}$ -small and D is a $\mathcal{U}$ -category, then $\text{Fun}(C, D)$ is a $\mathcal{U}$ -category.

Remark (1.1.2). Let D be a category satisfying the following properties:

- (C1) the set $\text{Ob}(D)$ is contained in the universe $\mathcal{U}$;
- (C2) for every pair (x, y) of objects of D , the set $\text{Hom}_D(x, y)$ is an element of $\mathcal{U}$.

The usual categories constructed from a universe $\mathcal{U}$ have these two properties: $\mathcal{U}\text{-Set}$, $\mathcal{U}\text{-Ab}$, and so on. Let C be a category belonging to $\mathcal{U}$. Then the category $\text{Fun}(C, D)$ does not in general have properties (C1) and (C2). For example, the category $\text{Fun}(C, \mathcal{U}\text{-Set})$ has neither property (C1) nor property (C2). This justifies the definition adopted for a $\mathcal{U}$ -category, rather than the more restrictive notion given by conditions (C1) and (C2) above.

Definition (1.2). Let C be a category. The category of presheaves of sets on C relative to the universe $\mathcal{U}$ —or, when no confusion can result, the category of presheaves on C —is the category of contravariant functors on C with values in the category of $\mathcal{U}$ -sets.

We denote by $\widehat{C}_{\mathcal{U}}$, or more simply by $\widehat{C}$ when no confusion can result, the category of presheaves of sets on C relative to the universe $\mathcal{U}$. The objects of $\widehat{C}_{\mathcal{U}}$ are called $\mathcal{U}$ -presheaves, or simply presheaves, on C . When C is $\mathcal{U}$ -small, the category $\widehat{C}_{\mathcal{U}}$ is a $\mathcal{U}$ -category. When C is a $\mathcal{U}$ -category, $\widehat{C}_{\mathcal{U}}$ is not necessarily a $\mathcal{U}$ -category.

Construction–Definition (1.3). Let x be an object of a $\mathcal{U}$ -category C . The $\mathcal{U}$ -functor represented by x is the functor

$$h_{\mathcal{U}}(x) : C^{\text{op}} \longrightarrow \mathcal{U}\text{-Set}$$

constructed as follows.⁴ Let y be an object of C .

³A category C is viewed as a set of arrows, equipped with the subset of objects, viewed as the set of identity arrows, and with the source and target maps. Thus the expressions ‘ C is an element of $\mathcal{U}$ ’ and ‘ C is $\mathcal{U}$ -small’ make sense.

⁴ C^{op} will always denote the category opposite to C .

(a) If $\text{Hom}_C(y, x)$ is an element of $\mathcal{U}$, then

$$h_{\mathcal{U}}(x)(y) = \text{Hom}_C(y, x).$$

(b) Suppose that $\text{Hom}_C(y, x)$ is not an element of $\mathcal{U}$, and let $R(Z, x, y)$ be the relation: ‘The set Z is the target of an isomorphism $\text{Hom}_C(y, x) \xrightarrow{\sim} Z$.’ We then put

$$h_{\mathcal{U}}(x)(y) = \tau_Z R(Z).$$

By axiom $(\mathcal{U}B)$, $h_{\mathcal{U}}(x)(y)$ is an element of $\mathcal{U}$. Let $R'(u, x, y)$ be the relation: ‘ u is a bijection

$$u : \text{Hom}_C(y, x) \xrightarrow{\sim} h_{\mathcal{U}}(x)(y)'.$$

We put

$$\varphi(y, x) = \tau_u R'(u).$$

Notice that in both cases (a) and (b) we have a canonical isomorphism

$$\varphi(y, x) : \text{Hom}_C(y, x) \xrightarrow{\sim} h_{\mathcal{U}}(x)(y).$$

In case (a), $\varphi(y, x)$ is the identity. Now let $u : y \rightarrow y'$ be an arrow of C . Composition of morphisms defines a map

$$\text{Hom}_C(u, x) : \text{Hom}_C(y', x) \longrightarrow \text{Hom}_C(y, x).$$

We set

$$h_{\mathcal{U}}(x)(u) = \varphi(y, x) \text{Hom}_C(u, x) \varphi(y', x)^{-1}.$$

One verifies immediately that $h_{\mathcal{U}}(x)$, so defined, is a functor $C^{\text{op}} \rightarrow \mathcal{U}\text{-Set}$.

1.3.1

Similarly, using the isomorphisms φ , for every morphism $v : x \rightarrow x'$ one defines a morphism of functors

$$h_{\mathcal{U}}(v) : h_{\mathcal{U}}(x) \longrightarrow h_{\mathcal{U}}(x'),$$

and one verifies immediately that this defines a functor

$$h_{\mathcal{U}} : C \longrightarrow \hat{C}_{\mathcal{U}}.$$

1.3.2

Let $\mathcal{V}$ now be a universe such that $\mathcal{U} \subset \mathcal{V}$. We then have a canonical fully faithful functor

$$\mathcal{U}\text{-Set} \hookrightarrow \mathcal{V}\text{-Set},$$

and hence a fully faithful functor

$$\hat{C}_{\mathcal{U}} \hookrightarrow \hat{C}_{\mathcal{V}}.$$

The diagram

$$\begin{array}{ccc} C & \xrightarrow{h_{\mathcal{U}}} & \hat{C}_{\mathcal{U}} \\ & \searrow h_{\mathcal{V}} & \downarrow \\ & & \hat{C}_{\mathcal{V}} \end{array}$$

commutes up to canonical isomorphism. When C is an element of $\mathcal{U}$, this canonical isomorphism is the identity.

1.3.3

In practice, the universe $\mathcal{U}$ is fixed once and for all and is not mentioned. We then use the notation $\widehat{C}$ for the category of $\mathcal{U}$ -presheaves of sets and

$$h : C \longrightarrow \widehat{C}.$$

For every object x of C , the presheaf $h(x)$ is called the *presheaf represented by x* , and we shall always identify its value at an object $y \in \text{Ob}(C)$ with $\text{Hom}_C(y, x)$.

Proposition (1.4). Let C be a $\mathcal{U}$ -category, let F be a presheaf on C , and let X be an object of C . There is an isomorphism, functorial in X and in F ,

$$i : \text{Hom}_{\widehat{C}}(h(X), F) \xleftarrow{\sim} F(X),$$

where, when F is of the form $h(Y)$, the map i is just the map

$$h : \text{Hom}(h(X), h(Y)) \longleftarrow \text{Hom}(X, Y).$$

In particular, the functor h is fully faithful.

1.4.1

This proposition justifies the usual abuse of language that identifies an object of C with the corresponding contravariant functor. A presheaf isomorphic to an object in the image of h , or, using the above abuse of language, isomorphic to an object of C , is called a *representable presheaf*.

1.4.2

Let $\mathcal{V}$ be a universe containing a universe $\mathcal{U}$. The category of $\mathcal{U}$ -presheaves is a *full* subcategory of the category of $\mathcal{V}$ -presheaves; consequently, a $\mathcal{U}$ -presheaf is representable if and only if its image in the category of $\mathcal{V}$ -presheaves is a representable $\mathcal{V}$ -presheaf.

Continuation anchor. The next untranslated source section is Exposé I, Section 2, ‘Projective and inductive limits’.

Exposé I. Presheaves

Sections 2–5

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Translator's status note. This is a working English translation of SGA 4, Exposé I, Sections 2–5, continuing the preceding translated fragment through Section 1.4. Obvious typographical slips in the working French T_EX source have been corrected where the mathematics fixes the reading.

2 Projective and inductive limits

Let C be a $\mathcal{U}$ -category and let I be a small category. Write $\text{Fun}(I, C)$ for the category of functors from I to C . The category $\text{Fun}(I, C)$ is a $\mathcal{U}$ -category. To every object X of C associate the subcategory $\mathcal{X}$ of C having X as its only object and the identity of X as its only arrow. Let $i_X : \mathcal{X} \rightarrow C$ be the inclusion functor. There is a unique functor $e_X : I \rightarrow \mathcal{X}$, and we shall denote by $k_X : I \rightarrow C$ the functor $i_X \circ e_X$. We shall say that k_X is the constant functor with value X . The correspondence $X \mapsto k_X$ is visibly functorial in X , which enables us to define, for every functor $G : I \rightarrow C$, the presheaf

$$X \mapsto \text{Hom}_{\text{Fun}(I, C)}(k_X, G).$$

Definition 2.1. The *projective limit* of G , denoted $\varprojlim_I G$, is the presheaf

$$X \mapsto \text{Hom}_{\text{Fun}(I, C)}(k_X, G).$$

When the presheaf $\varprojlim_I G$ is representable, we again denote by $\varprojlim_I G$ an object of C representing it. The object $\varprojlim_I G$ is therefore defined only up to isomorphism. When no confusion can result, we use the abbreviated notation $\varprojlim G$.

As G varies, $\varprojlim G$ is a functor from $\text{Fun}(I, C)$ to $\widehat{C}$.

2.1.1

By symmetry, reversing the direction of the arrows in C , one defines similarly the inductive limit of a functor: it is a covariant functor on C with values in the category of $\mathcal{U}$ -sets. We shall use the notations $\varinjlim_I G$ and $\varinjlim G$.

Products, fiber products, and kernels are projective limits. Similarly, sums, amalgamated sums, and cokernels are inductive limits.

Definition 2.2. Let I and C be two categories, and let $G : I \rightarrow C$ be a functor. We say that the projective limit of G is *representable* if there exists a universe $\mathcal{U}$ such that:

- 1) the category I is $\mathcal{U}$ -small;
- 2) the category C is a $\mathcal{U}$ -category;
- 3) the presheaf $\varprojlim G$, with values in the category of $\mathcal{U}$ -sets, is representable.

It follows from no. 1 that the object $\varprojlim G$ representing the presheaf $\varprojlim G$ does not depend, up to isomorphism, on the universe $\mathcal{U}$. Note also that there is a smallest universe $\mathcal{U}'$ having properties (1) and (2), and that the presheaf $\varprojlim G$ necessarily takes values in the category of $\mathcal{U}'$ -sets.

2.2.1

Let I and C be two categories. We say that *I -projective limits in C are representable* if, for every functor $G : I \rightarrow C$, the projective limit of G is representable. Finally, let C be a category and let $\mathcal{U}$ be a universe. We say that *$\mathcal{U}$ -projective limits in C are representable* if, for every $\mathcal{U}$ -small category I and every functor $G : I \rightarrow C$, the projective limit of G is representable.

Proposition 2.3. Let C be a category and let $\mathcal{U}$ be a universe. The following assertions are equivalent:

- i) $\mathcal{U}$ -projective limits in C are representable.
- ii) Products indexed by a small set are representable, and kernels of pairs of arrows are representable.
- iii) Products indexed by a small set are representable, and fiber products are representable.

Proof. It suffices to observe that there is an isomorphism, functorial in G ,

$$\varprojlim G = \text{Ker} \left(\prod_{i \in \text{Ob}(I)} G(i) \rightrightarrows \prod_{u \in \text{Fl}(I)} G(\text{target}(u)) \right),$$

where the pair of arrows is defined by the morphisms

$$\begin{aligned} \prod_{i \in \text{Ob}(I)} G(i) &\xrightarrow{\text{pr}_{\text{target}(u)}} G(\text{target}(u)), \\ \prod_{i \in \text{Ob}(I)} G(i) &\xrightarrow{\text{pr}_{\text{source}(u)}} G(\text{source}(u)) \xrightarrow{G(u)} G(\text{target}(u)). \end{aligned}$$

Moreover, it is clear that

$$\text{Ker} \left(X \begin{smallmatrix} \xrightarrow{u} \\ \xrightarrow{v} \end{smallmatrix} Y \right) = X \times_{X \times Y} X,$$

where the two morphisms from X to $X \times Y$ are $\text{id}_X \times u$ and $\text{id}_X \times v$.

2.3.1

There are, of course, analogous definitions and assertions for inductive limits, which we shall not spell out. The same applies to finite projective and inductive limits, that is, those relative to a finite category I .

Corollary 2.3.2. Let $\mathcal{U}\text{-Set}$ denote the category of $\mathcal{U}$ -sets, and let $\mathcal{U}\text{-Ab}$ denote the category of abelian-group objects of $\mathcal{U}\text{-Set}$. The $\mathcal{U}$ -projective and $\mathcal{U}$ -inductive limits in $\mathcal{U}\text{-Set}$ and in $\mathcal{U}\text{-Ab}$ are representable.

Proposition 2.3.3. Let I be a small category, let $F : I \rightarrow \mathcal{U}\text{-Set}$ be a functor, and let $G \subset \text{Ob}(I)$ be a small set of objects of I such that, for every object X of I , there exists an object $Y \in G$ and a morphism $Y \rightarrow X$ (respectively $X \rightarrow Y$). Then $\varprojlim_I F$ (respectively $\varinjlim_I F$) is representable, and

$$\text{card}(\varprojlim_I F) \leq \prod_{Y \in G} \text{card}(F(Y))$$

respectively

$$\text{card}(\varprojlim_I F) \leq \sum_{Y \in G} \text{card}(F(Y)).$$

Proof. One checks immediately that $\varprojlim_I F$ (respectively $\varinjlim_I F$) is isomorphic to a subobject of $\prod_{Y \in G} F(Y)$ (respectively to a quotient of $\coprod_{Y \in G} F(Y)$), whence the proposition.

Definition 2.4.1. Let C be a category in which finite projective (respectively inductive) limits are representable, and let $u : C \rightarrow C'$ be a functor. We say that u is *left exact* (respectively *right exact*) if it “commutes” with finite projective (respectively finite inductive) limits. A functor that is both left and right exact is called *exact*.

2.4.2

It follows from 2.3 that, for a functor to be left exact, it is necessary and sufficient that it carry the final object, i.e. the empty product, to the final object, the product of two objects to the product of their images, and the kernel of a pair of arrows to the kernel of the image pair; equivalently, that it carry the final object to the final object and fiber products to fiber products. Here it is assumed that finite projective limits in C are representable.

2.5.0

Let I , J , and C be three categories, and let $G : I \times J \rightarrow C$ be a functor, i.e. a functor from I with values in the category of functors from J to C . Suppose that the projective limits of the functors

$$\begin{aligned} G_i : J &\longrightarrow C & (i \in \text{Ob}(I)) \\ j &\longmapsto G(i, j) \end{aligned}$$

are representable, and that the functor

$$i \longmapsto \varprojlim_J G_i$$

admits a representable projective limit. Then it is clear that the functor G admits a representable projective limit and that there is a canonical isomorphism

$$\varprojlim_{I \times J} G \xrightarrow{\sim} \varprojlim_I \varprojlim_J G_i,$$

and hence a canonical isomorphism

$$\varprojlim_I \varprojlim_J G_i \xrightarrow{\sim} \varprojlim_J \varprojlim_I G_j.$$

We shall say more briefly that projective limits commute with projective limits. One sees in the same way that inductive limits commute with inductive limits. But it is not true in general that inductive limits commute with projective limits.

Definition 2.5. Let C be a category with representable fiber products, let I be a category, let $G : I \rightarrow C$ be a functor, let $g : G \rightarrow X$ be a morphism from G to an object of C , i.e. a morphism from G to the constant functor associated with X , and let $m : Y \rightarrow X$ be a morphism of C . Let

$$G \times_X Y : I \longrightarrow C$$

be the functor

$$i \longmapsto G(i) \times_X Y.$$

We say that the inductive limit of G is *universal* if, for every object X , every morphism $g : G \rightarrow X$, and every morphism $m : Y \rightarrow X$,

- a) the inductive limit of the functor $G \times_X Y$ is representable;
- b) the canonical morphism

$$\varinjlim (G \times_X Y) \longrightarrow (\varinjlim G) \times_X Y$$

is an isomorphism.

Proposition 2.6. Let $\mathcal{U}$ be a universe. Inductive limits in $\mathcal{U}\text{-Set}$ that are representable, in particular $\mathcal{U}$ -inductive limits (2.2.1) in $\mathcal{U}\text{-Set}$, are universal.

We shall also use another commutation result between projective and inductive limits, which we now present.

Definition 2.7. A category I is *pseudo-filtered* if it has the following properties:

PS 1) Every diagram of the form

$$\begin{array}{ccc} & & j \\ & \nearrow & \\ i & & \\ & \searrow & \\ & & j' \end{array}$$

can be inserted into a commutative diagram

$$\begin{array}{ccccc} & & j & & \\ & \nearrow & & \searrow & \\ i & & & & k \\ & \searrow & & \nearrow & \\ & & j' & & \end{array} .$$

PS 2) Every diagram of the form

$$i \begin{array}{c} \xrightarrow{u} \\ \xrightarrow{v} \end{array} j$$

can be inserted into a diagram

$$i \begin{array}{c} \xrightarrow{u} \\ \xrightarrow{v} \end{array} j \xrightarrow{w} k$$

such that $w \circ u = w \circ v$.

A category I is called *filtered* if it is pseudo-filtered, nonempty, and connected, i.e. if any two objects of I can be joined by a finite chain of arrows, with no condition imposed on the directions of the arrows. Equivalently, in the presence of PS 2, this means that $I \neq \emptyset$ and that, for any two objects a, b of I , there is always an object c of I and arrows $a \rightarrow c$ and $b \rightarrow c$. A category I is called *cofiltered* if I^{op} is filtered.

Example 2.7.1. If, in I , amalgamated sums (respectively sums of two objects) and cokernels of double arrows are representable, then I is pseudo-filtered (respectively filtered).

Proposition 2.8. Let $\mathcal{U}$ be a universe. Filtered $\mathcal{U}$ -inductive limits in $\mathcal{U}\text{-Set}$ commute with finite projective limits.

One is immediately reduced to proving that filtered $\mathcal{U}$ -inductive limits commute with fiber products. The proof is left to the reader. One may use the following description of the limit.

Lemma 2.8.1. Let I be a small filtered category, and let $i \mapsto X_i$ be a functor from I to $\mathcal{U}\text{-Set}$. On the sum set $\coprod_{i \in \text{Ob } I} X_i$, let R be the following relation:

- (R) Two elements $x_i \in X_i$ and $x_j \in X_j$ are related if there exist an object $k \in \text{Ob } I$ and two morphisms $u : i \rightarrow k$ and $v : j \rightarrow k$ such that the images in X_k of x_i and x_j under the transition maps u and v , respectively, are equal.

Then:

- 1) R is an equivalence relation.
- 2) The quotient $\coprod_{i \in \text{Ob } I} X_i / R$ is canonically isomorphic to $\varinjlim_I X_i$.
- 3) Two elements $x_i \in X_i$ and $x_j \in X_j$ are respectively equivalent, modulo R , to two elements of one and the same X_k .
- 4) For every $i \in \text{Ob } I$, two elements α and β of X_i are equivalent modulo R if and only if there exists a morphism $u : i \rightarrow j$ such that $u(\alpha) = u(\beta)$.

Assertion 1) follows from PS 1 (2.7). To prove 2), one checks that $\coprod_{i \in \text{Ob } I} X_i / R$ has the universal property of the inductive limit; here only PS 1 is used. Assertion 3) follows from the fact that I is connected. Assertion 4) follows from PS 2.

Corollary 2.9. Let γ be a species of algebraic structure “defined by finite projective limits.” The reader is asked to give a mathematical meaning to the preceding phrase; let us merely note that the structures of groups, abelian groups, rings, modules, etc. are structures of this kind. Let $\mathcal{U}\text{-}\gamma$ denote the category of γ -objects of $\mathcal{U}\text{-Set}$. The functor that associates with every object of $\mathcal{U}\text{-}\gamma$ its underlying set commutes with filtered $\mathcal{U}$ -inductive limits. Consequently filtered $\mathcal{U}$ -inductive limits in $\mathcal{U}\text{-}\gamma$ commute with finite projective limits.

Corollary 2.10. Pseudo-filtered $\mathcal{U}$ -inductive limits in $\mathcal{U}\text{-Ab}$ commute with finite projective limits.

One is reduced to filtered limits by decomposing the indexing category into its connected components.

We now record a result that we shall use constantly. Let C and C' be two $\mathcal{U}$ -categories, and let

$$C \begin{array}{c} \xrightarrow{u} \\ \xleftarrow{v} \end{array} C'$$

be two functors, with u *left adjoint* to v . Recall that this means that there is an isomorphism of bifunctors with values in $\mathcal{U}\text{-Set}$,

$$\mathrm{Hom}_{C'}(u(X), X') \xrightarrow{\sim} \mathrm{Hom}_C(X, v(X')).$$

Proposition 2.11. The functor u commutes with representable inductive limits; the functor v commutes with representable projective limits.

This assertion means that, for every category I and every functor $G : I \rightarrow C'$ such that the projective limit of G is representable, the functor $v \circ G$ admits a representable projective limit and there is a canonical isomorphism

$$v(\varprojlim G) \longrightarrow \varprojlim (v \circ G).$$

The reader will write out for himself the assertion concerning the functor u .

Finally, let us point out a computation of projective limits in the case where the indexing category admits products.

Proposition 2.12. Let I and C be two categories, and let $\mathcal{V}$ be a universe. Suppose that $\mathcal{V}$ -projective limits in C are representable and that there exists in I a family of objects $(i_\alpha)_{\alpha \in A}$, with $A \in \mathcal{V}$, such that:

- 1) the products $i_\alpha \times i_\beta$ are representable for every pair $(\alpha, \beta) \in A \times A$;
- 2) every object of I maps to at least one of the i_α .

Then, for every contravariant functor

$$F : I^{\mathrm{op}} \longrightarrow C,$$

the projective limit of F exists and there is an isomorphism, functorial in F ,

$$\varprojlim F \xrightarrow{\sim} \mathrm{Ker} \left(\prod_{\alpha \in A} F(i_\alpha) \rightrightarrows \prod_{(\alpha, \beta) \in A \times A} F(i_\alpha \times i_\beta) \right),$$

where the two arrows are defined by the projections of the products $i_\alpha \times i_\beta$ onto the factors.

3 Exactness properties of the category of presheaves

Let C be a $\mathcal{U}$ -category, and let $\widehat{C}$ be the category of presheaves on C . The exactness properties of $\widehat{C}$ are all deduced from the following proposition.

Proposition 3.1. The $\mathcal{U}$ -projective and $\mathcal{U}$ -inductive limits in $\widehat{C}$ are representable. For every object X of C , the functor on $\widehat{C}$

$$F \longmapsto F(X), \quad F \in \widehat{C},$$

commutes with inductive and projective limits.

In other words, inductive and projective limits in $\widehat{C}$ are computed argument by argument. We list a few corollaries.

Corollary 3.2. Let γ be an algebraic structure defined by finite projective limits. The category of contravariant functors with values in $\mathcal{U}\text{-}\gamma$ is equivalent to the category of γ -objects of $\widehat{C}$.

Corollary 3.3. A morphism of $\widehat{C}$ that is both a monomorphism and an epimorphism is an isomorphism. A morphism of $\widehat{C}$ factors uniquely as an epimorphism followed by a monomorphism. Representable inductive limits in $\widehat{C}$ are universal. Filtered $\mathcal{U}$ -inductive limits in $\widehat{C}$ commute with finite projective limits. The canonical functor $C \rightarrow \widehat{C}$ commutes with representable projective limits.

More generally, one may say that the category $\widehat{C}$ inherits all the properties of the category of $\mathcal{U}$ -sets that involve inductive and projective limits.

3.4.0

Let F be an object of $\widehat{C}$. We denote by C/F the following category. Its objects are pairs consisting of an object X of C and a morphism u from X to F . Given two objects (X, u) and (Y, v) , a morphism from (X, u) to (Y, v) is a morphism $g : X \rightarrow Y$ such that the following diagram is commutative:

$$\begin{array}{ccc} X & \xrightarrow{g} & Y \\ & \searrow u & \swarrow v \\ & F & \end{array}$$

Proposition 3.4. With the notation of (3.4.0), the source functor $C/F \rightarrow \widehat{C}$ admits a representable inductive limit in $\widehat{C}$. The canonical morphism

$$\varinjlim_{C/F} \text{source}(\cdot) \longrightarrow F$$

is an isomorphism.

Corollary 3.5. Let F and H be two objects of $\widehat{C}$. There is a canonical isomorphism

$$\text{Hom}(F, H) \xrightarrow{\sim} \varprojlim_{(X, u) \in \text{Ob}(C/F)} H(X).$$

Proof. The corollary follows immediately from 3.4 and 1.2.

Proposition 3.6. Let C be a $\mathcal{U}$ -category, let $\mathcal{V}$ be a universe containing $\mathcal{U}$, and let

$$i : \widehat{C}_{\mathcal{U}} \longrightarrow \widehat{C}_{\mathcal{V}}$$

be the natural inclusion functor between the corresponding categories of presheaves (1.3). The functor i commutes with inductive and projective limits. Moreover, for every object F of $\widehat{C}_{\mathcal{U}}$, every subobject of $i(F)$ is isomorphic to the image under i of a unique subobject of F ; thus one obtains a bijection from the set of subobjects of F to the set of subobjects of $i(F)$.

4 Sieves

Definition 4.1. Let C be a category. A *sieve* of the category C is a full subcategory D of C having the following property: every object of C admitting a morphism to an object of D belongs to D . If X is an object of C , the sieves of the category C/X will be called, by abuse of language, *sieves of X* .

Let $\mathcal{U}$ be a universe such that C is a $\mathcal{U}$ -category, and let $\widehat{C}$ be the corresponding category of presheaves. To every sieve of X one associates a subobject of X in $\widehat{C}$ as follows: to every object Y of C one assigns the set of morphisms $f : Y \rightarrow X$ such that the object (Y, f) belongs to the sieve.

Proposition 4.2. The map defined above establishes a bijection between the set of sieves of X and the set of subobjects of X in $\widehat{C}$.

Proof. We only indicate the inverse map. To every subfunctor R of X one associates the category C/R of objects of C over R (3.4). One checks immediately that C/R is a sieve of X .

Remark 4.2.1. One sees in the same way that the sieves of C are in canonical one-to-one correspondence with the set of subfunctors of the final functor on C , the final object of $\widehat{C}$.

4.3

Let C be a $\mathcal{U}$ -category. By abuse of language, we shall also call the subobjects of X in the category $\widehat{C}$ the sieves of X . This abuse of language allows us, for every presheaf F and every sieve R of X , to define $\text{Hom}_{\widehat{C}}(R, F)$ as the set of morphisms from the functor R to F . There is, moreover, an isomorphism canonical and functorial in F (3.5):

$$\text{Hom}_{\widehat{C}}(R, F) \xrightarrow{\sim} \varprojlim_{C/R} F(.),$$

which also gives a direct definition.¹ Similarly, Proposition 4.2 allows us to transpose the usual operations on functors to sieves. We mention the following.

4.3.1 Base change

Let R be a sieve of X , and let $f : Y \rightarrow X$ be a morphism of objects of C . The fiber product $R \times_X Y$ is a sieve of Y , called the *sieve deduced from R by base change*. The corresponding subcategory of C/Y is the inverse image of the subcategory of C/X defined by R , under the canonical functor $C/Y \rightarrow C/X$ defined by f .

4.3.2 Order relation, intersection, union

The inclusion relation on subfunctors of X is an order relation. One may define the union and intersection of a family of sieves indexed by an arbitrary set as the least upper bound and greatest lower bound of the corresponding family of subpresheaves.

¹More simply, C/R identifies with R , so that one has the formula $\text{Hom}_{\widehat{C}}(R, F) \xrightarrow{\sim} \varprojlim_R F$, where F abusively denotes the composite of F with the tautological source functor $R \rightarrow C/X \rightarrow C$.

4.3.3 Image, generated sieve

Let $(F_\alpha)_{\alpha \in A}$ be a family of presheaves, and for each $\alpha \in A$ let $f_\alpha : F_\alpha \rightarrow X$ be a morphism, where X is an object of C . The *image* of this family of morphisms is the union of the images of the f_α . The image of this family is therefore a sieve of X . In particular, if all the F_α are objects of C , the image sieve will be called the sieve generated by the morphisms f_α . The category C/R is the full subcategory of C/X formed by those objects $X' \rightarrow X$ over X for which there exists an X -morphism from X' to one of the F_α .

As an exercise, the reader may translate the relations and operations just defined on subfunctors into terms of the categories C/R . He will then see that these relations and operations do not depend on the universe $\mathcal{U}$ such that C is a $\mathcal{U}$ -category, and that they are therefore defined for every category, without requiring one to specify the universe to which the sets of morphisms belong. This was in any case already foreseeable *a priori* from 3.6.

5 Functoriality of categories of presheaves

5.0

Let C , C' , and D be three categories, and let $u : C \rightarrow C'$ be a functor. We denote by u^* the functor obtained by composition with u :

$$u^* : \text{Hom}((C')^{\text{op}}, D) \longrightarrow \text{Hom}(C^{\text{op}}, D), \quad G \longmapsto G \circ u.$$

The functor u^* commutes with inductive and projective limits.

Proposition 5.1. Suppose that C is small and that, in D , $\mathcal{U}$ -inductive limits (respectively projective limits) are representable. The functor u^* admits a left adjoint $u_!$ (respectively a right adjoint u_*). Thus one has an isomorphism

$$\begin{aligned} \text{Hom}_{\text{Hom}(C^{\text{op}}, D)}(F, u^*G) &\xrightarrow{\sim} \text{Hom}_{\text{Hom}((C')^{\text{op}}, D)}(u_!F, G), \\ (\text{respectively } \text{Hom}_{\text{Hom}(C^{\text{op}}, D)}(u^*G, F) &\simeq \text{Hom}_{\text{Hom}((C')^{\text{op}}, D)}(G, u_*F)). \end{aligned}$$

Proof. We indicate only the proof of the existence of the left adjoint. The corresponding part of the proposition follows formally from the isomorphisms

$$\text{Hom}(C^{\text{op}}, D) \xrightarrow{\sim} \text{Hom}(C, D^{\text{op}})^{\text{op}} \xrightarrow{\sim} \text{Hom}((C^{\text{op}})^{\text{op}}, D^{\text{op}})^{\text{op}}.$$

Let Y be an object of C' . Denote by I_u^Y the following category. Its objects are pairs (X, m) , where X is an object of C and m is a morphism $Y \rightarrow u(X)$. If (X, m) and (X', m') are two objects of I_u^Y , a morphism from (X, m) to (X', m') is a morphism $\xi : X \rightarrow X'$ such that $m' = u(\xi)m$. Composition is defined in the evident way.

Let $f : Y \rightarrow Y'$ be a morphism of C' . By composition, f defines a functor $I_u^f : I_u^Y \rightarrow I_u^{Y'}$.

We also have a functor $\text{pr}_Y : I_u^Y \rightarrow C$ which sends the object (X, m) to the object X . Then the evident diagram

$$\begin{array}{ccc} I_u^Y & \xrightarrow{I_u^f} & I_u^{Y'} \\ \text{pr}_Y \downarrow & \swarrow \text{pr}_{Y'} & \\ C & & \end{array} \quad (*)$$

is commutative.

Now let F be a presheaf on C and set

$$u_!F(Y) = \varinjlim_{I_u^Y} F \circ \text{pr}_Y. \quad (5.1.1)$$

The commutativity of diagram (*) and the functoriality of inductive limits make $u_!F$ into a presheaf on C' . Let us show that the functor $u_!$ is left adjoint to u^* . For this it is enough to show that, for every presheaf G on C' , there is a functorial isomorphism

$$\text{Hom}(u_!F, G) \xrightarrow{\sim} \text{Hom}(F, u^*G).$$

Let $\xi \in \text{Hom}(u_!F, G)$. For every object X of C , we have a morphism

$$\xi_{u(X)} : u_!F(u(X)) \longrightarrow G(u(X)).$$

But $(X, \text{id}_{u(X)})$ is an object of $I_u^{u(X)}$, and by the definition of the inductive limit there is a canonical morphism

$$F(X) \longrightarrow u_!F(u(X)).$$

We obtain, for every object X of C , a morphism

$$\eta_X : F(X) \longrightarrow G(u(X)),$$

which is visibly functorial in X . Hence a morphism

$$\eta : F \longrightarrow u^*G.$$

Conversely, let $\eta \in \text{Hom}(F, u^*G)$. One obtains, for every object Y of C' , a morphism of functors

$$\eta_Y : F \circ \text{pr}_Y \longrightarrow (u^*G) \circ \text{pr}_Y,$$

and hence, by composing with the evident morphism from the functor $(u^*G) \circ \text{pr}_Y$ to the constant functor $G(Y)$, a morphism

$$\xi_Y : u_!F(Y) \longrightarrow G(Y),$$

which is functorial in Y . Hence a morphism

$$\xi : u_!F \longrightarrow G.$$

The reader will check that the two maps thus defined are inverse to one another, completing the proof.

Proposition 5.2. Suppose that in D the $\mathcal{U}$ -inductive limits are representable, the finite projective limits are representable, and filtered $\mathcal{U}$ -inductive limits commute with finite projective limits. Suppose moreover that, in C , finite projective limits are representable and that the functor u is left exact (2.3.2). Then finite projective limits are representable in $\text{Hom}(C^{\text{op}}, D)$ and in $\text{Hom}((C')^{\text{op}}, D)$, and the functor $u_!$ is left exact.

Proof. The first assertion is trivial. Let us prove the second. By the proof of 5.1, for every presheaf F on C and every object Y of C' one has

$$u_!F(Y) \xrightarrow{\sim} \varinjlim_{I_u^Y} F \circ \text{pr}_Y.$$

It is therefore enough to show that the category $(I_u^Y)^{\text{op}}$ satisfies axioms PS 1 and PS 2 (2.7), and that this category is connected. The verification is left to the reader.

5.3

Specializing these results to the case where D is the category of $\mathcal{U}$ -sets, one obtains a sequence of three functors

$$u_!, \quad u^*, \quad u_*,$$

which is a “sequence of adjoint functors” in the sense that, for two consecutive functors in the sequence, the one on the right is right adjoint to the other. Their essential properties are summarized in the following proposition.

Proposition 5.4. Let C be a small category, let C' be a $\mathcal{U}$ -category, and let $u : C \rightarrow C'$ be a functor.

- 1) The functor $u^* : \widehat{C'} \rightarrow \widehat{C}$ commutes with inductive and projective limits.
- 2) The functor $u_* : \widehat{C} \rightarrow \widehat{C'}$ commutes with projective limits. For every presheaf F on C and every object Y of C' , one has

$$u_* F(Y) \xrightarrow{\sim} \text{Hom}_{\widehat{C}}(u^*(Y), F).$$

- 3) The functor $u_! : \widehat{C} \rightarrow \widehat{C'}$ commutes with inductive limits. The functor $u_!$ is defined only up to isomorphism, but it can always be chosen so that the diagram

$$\begin{array}{ccc} C & \xrightarrow{u} & C' \\ h \downarrow & & \downarrow h' \\ \widehat{C} & \xrightarrow{u_!} & \widehat{C'} \end{array}$$

is commutative, where h and h' are the canonical inclusion functors.

For every presheaf F on C , one has

$$u_! F \xrightarrow{\sim} \varinjlim_{C/F} h' \circ u.$$

- 4) If finite projective limits are representable in C and if u is left exact (2.3.2), then the functor $u_!$ is left exact.

Proof. Assertion (1) is trivial. Assertion (2) follows from the fact that u_* is a right adjoint, by 2.11 and 1.4. The same applies to assertion (3), but one also uses 3.4. Finally, assertion (4) is precisely 5.2, which applies by 2.7.

Proposition 5.5. Let C and C' be two small categories, and let

$$C \xrightleftharpoons[v]{u} C'$$

be a pair of functors, where v is left adjoint to u . Then there exist isomorphisms, compatible with the adjunction isomorphisms,

$$v^* \xrightarrow{\sim} u_!, \quad v_* \xrightarrow{\sim} u^*.$$

Proof. It is enough to exhibit an isomorphism $v^* \xrightarrow{\sim} u_!$; the other isomorphism will follow by adjunction. Let F be a presheaf on C and let Y be an object of C' . Then

$$v^* F(Y) \xrightarrow{\sim} \text{Hom}(v(Y), F).$$

Using 3.4, we get

$$\text{Hom}(v(Y), F) \xrightarrow{\sim} \varinjlim_{C/F} \text{Hom}(v(Y), .).$$

Since v is left adjoint to u , it follows that

$$\varinjlim_{C/F} \text{Hom}(v(Y), .) \xrightarrow{\sim} \varinjlim_{C/F} \text{Hom}(Y, u(.)).$$

Using 5.4.3), we then obtain

$$\varinjlim_{C/F} \text{Hom}(Y, u(.)) \xrightarrow{\sim} \text{Hom}(Y, u_! F) \xrightarrow{\sim} u_! F(Y).$$

Thus, for every object Y of C' , we have determined an isomorphism $v^* F(Y) \xrightarrow{\sim} u_! F(Y)$, visibly functorial in Y and in F .

Corollary 5.5.1. Let $u : C \rightarrow C'$ be a functor that admits a left adjoint. The functor $u_! : \widehat{C} \rightarrow \widehat{C'}$ commutes with projective limits; recall that it commutes with inductive limits by 5.4.3.

Remark 5.5.2. One thus obtains a “sequence of *four* adjoint functors” (cf. 5.3):

$$v_!, \quad v^* = u_!, \quad v_* = u^*, \quad u_*,$$

of which the first three (respectively the last three) commute with inductive limits (respectively projective limits).

Proposition 5.6. The hypotheses are those of 5.4. The following conditions are equivalent:

- i) the functor u is fully faithful;
- ii) the functor $u_!$ is fully faithful;
- iii) the adjunction morphism $\text{id}_{\widehat{C}} \rightarrow u^* u_!$ is an isomorphism;
- iv) the functor u_* is fully faithful;
- v) the adjunction morphism $u^* u_* \rightarrow \text{id}_{\widehat{C}}$ is an isomorphism.

Proof. It is clear that ii) $\Leftrightarrow$ iii) and iv) $\Leftrightarrow$ v), by general properties of adjoint functors, and that ii) $\Rightarrow$ i), by 5.4.3). Let us prove that i) $\Rightarrow$ iii). The functors $\text{id}_{\widehat{C}}$, u^* , and $u_!$ commute with inductive limits. By 3.4, it is therefore enough to prove that $H \rightarrow u^* u_! H$ is an isomorphism when H is representable, which is evident.

It remains to show that iii) is equivalent to v). For every object H (respectively K) of $\widehat{C}$, denote by $\Phi(H) : H \rightarrow u^*u_!H$ (respectively by $\Psi(K) : u^*u_*K \rightarrow K$) the adjunction morphism. Then one has a commutative diagram

$$\begin{array}{ccc}
\mathrm{Hom}_{\widehat{C}}(H, u^*u_*K) & \xrightarrow{\mathrm{Hom}(H, \Psi(K))} & \mathrm{Hom}_{\widehat{C}}(H, K) \\
\downarrow \sim & & \nearrow \\
\mathrm{Hom}_{\widehat{C}'}(u_!H, u_*K) & & \mathrm{Hom}(\Phi(H), K) \\
\downarrow \sim & & \\
\mathrm{Hom}_{\widehat{C}}(u^*u_!H, K) & &
\end{array}$$

Hence $\Phi(H)$ is an isomorphism for every H if and only if $\Psi(K)$ is an isomorphism for every K .

Remark 5.7.

- a) The equivalences ii) $\Leftrightarrow$ iii) $\Leftrightarrow$ iv) $\Leftrightarrow$ v) are general facts about adjoint functors.
- b) The explicit form of $u_!$, respectively u_* , given in the proof of 5.1 shows immediately that, under the general hypotheses of 5.1, if u is fully faithful, then the adjunction morphism $\mathrm{id} \rightarrow u^*u_!$ (respectively $u^*u_* \rightarrow \mathrm{id}$) is an isomorphism; equivalently, $u_!$ (respectively u_*) is fully faithful.

5.8.0

Let γ be a species of algebraic structure defined by finite projective limits. Let $\mathcal{U}\text{-}\gamma\text{-}\mathbf{Set}$ be the category of γ -objects of $\mathcal{U}\text{-}\mathbf{Set}$, and let

$$\mathrm{und} : \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set} \longrightarrow \mathcal{U}\text{-}\mathbf{Set}$$

be the underlying-set functor. For simplicity, we assume that the species of structure under consideration has a single underlying set. Let C be a category. Composition with und gives a functor denoted

$$\widehat{\mathrm{und}} : \mathrm{Hom}(C^{\mathrm{op}}, \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set}) \longrightarrow \widehat{C}.$$

Since in $\widehat{C}$ projective limits are computed argument by argument, the functor $\widehat{\mathrm{und}}$ factors as an equivalence

$$\mathrm{Hom}(C^{\mathrm{op}}, \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set}) \xrightarrow{\sim} \widehat{C}_\gamma, \tag{5.8.1}$$

where $\widehat{C}_\gamma$ is the category of γ -objects of $\widehat{C}$, followed by a functor again denoted

$$\widehat{\mathrm{und}} : \widehat{C}_\gamma \longrightarrow \widehat{C},$$

and called the *underlying presheaf of sets* functor.

5.8.2

Suppose that the functor

$$\mathcal{U}\text{-}\gamma\text{-}\mathbf{Set} \longrightarrow \mathcal{U}\text{-}\mathbf{Set}$$

admits a left adjoint

$$\mathbf{Free} : \mathcal{U}\text{-}\mathbf{Set} \longrightarrow \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set},$$

the functor of “freely generated γ -objects.” Examples are the free group, free commutative group, free A -module, and so on. In fact one can show that this condition is always satisfied. Composition with $\mathbf{Free}$ gives a functor

$$\widehat{\mathbf{Free}} : \widehat{C} \longrightarrow \mathbf{Hom}(C^{\text{op}}, \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set}),$$

and, by composing with the equivalence (5.8.1), a functor again denoted

$$\widehat{\mathbf{Free}} : \widehat{C} \longrightarrow \widehat{C}_\gamma,$$

and called the functor “presheaf of freely generated γ -objects.” The functor $\widehat{\mathbf{Free}}$ is left adjoint to $\widehat{\text{und}}$.

Proposition 5.8.3. Let γ be a species of algebraic structure defined by finite projective limits such that, in the category of γ -objects of $\mathcal{U}\text{-}\mathbf{Set}$, $\mathcal{U}$ -inductive limits are representable.² Let C be a category belonging to $\mathcal{U}$, let C' be a $\mathcal{U}$ -category, and let $u : C \rightarrow C'$ be a functor. Denote by $\widehat{C}_\gamma$ (respectively $\widehat{C}'_\gamma$) the category of γ -objects of $\widehat{C}$ (respectively $\widehat{C}'$), and by $(u^*)^\gamma$ the functor on γ -objects deduced from u^* . It follows from 5.1 and the equivalence 5.8.1 that there exists a functor left adjoint (respectively right adjoint) to $(u^*)^\gamma$. This functor is denoted $u_{!\gamma}$ (respectively $u_{*\gamma}$).

- 1) The functor $(u^*)^\gamma$ commutes with inductive and projective limits. The diagram

$$\begin{array}{ccc} \widehat{C}'_\gamma & \xrightarrow{(u^*)^\gamma} & \widehat{C}_\gamma \\ \widehat{\text{und}}' \downarrow & & \downarrow \widehat{\text{und}} \\ \widehat{C}' & \xrightarrow{u^*} & \widehat{C} \end{array} \quad (*)$$

is commutative; here $\widehat{\text{und}}'$ and $\widehat{\text{und}}$ denote the underlying-set functors.

- 2) The functor $u_{*\gamma}$ commutes with projective limits. The diagram

$$\begin{array}{ccc} \widehat{C}_\gamma & \xrightarrow{u_{*\gamma}} & \widehat{C}'_\gamma \\ \widehat{\text{und}} \downarrow & & \downarrow \widehat{\text{und}}' \\ \widehat{C} & \xrightarrow{u_*} & \widehat{C}' \end{array} \quad (**)$$

is commutative up to isomorphism.

²One can show that this condition is always satisfied.

- 3) The functor $u_{! \gamma}$ commutes with inductive limits. Suppose that $\widehat{\text{und}}$ (respectively $\widehat{\text{und}}'$) admits a left adjoint $\widehat{\text{Free}}$ (respectively $\widehat{\text{Free}}'$), as in (5.8.2). The diagram

$$\begin{array}{ccc}
 \widehat{C} & \xrightarrow{u_{!}} & \widehat{C}' \\
 \widehat{\text{Free}} \downarrow & & \downarrow \widehat{\text{Free}}' \\
 \widehat{C}_{\gamma} & \xrightarrow{u_{! \gamma}} & \widehat{C}'_{\gamma}
 \end{array} \tag{***}$$

is commutative up to isomorphism.

Suppose that $u_{!}$ commutes with finite projective limits (5.4 and 5.6). Then the diagram

$$\begin{array}{ccc}
 \widehat{C}_{\gamma} & \xrightarrow{u_{! \gamma}} & \widehat{C}'_{\gamma} \\
 \widehat{\text{und}} \downarrow & & \downarrow \widehat{\text{und}}' \\
 \widehat{C} & \xrightarrow{u_{!}} & \widehat{C}'
 \end{array} \tag{****}$$

is commutative up to isomorphism, and $u_{! \gamma}$ commutes with finite projective limits.

Proof. Assertion (1) is evident. Assertion (2) is also evident, since u_{*} is a right adjoint and hence, by 2.11, commutes with projective limits and in particular with finite projective limits; this gives the commutativity of diagram (**). The commutativity of diagram (***) follows from the uniqueness, up to isomorphism, of the left adjoint functor. Finally, the commutativity of diagram (****) follows immediately from the fact that $u_{!}$ commutes with finite projective limits.

Notation 5.9. By abuse of notation, the functors $(u^{*})^{\gamma}$ and $u_{* \gamma}$ will often be denoted by u^{*} and u_{*} ; this risks no confusion in view of the commutativity of diagrams (*) and (**). In contrast, when $u_{!}$ does not commute with finite projective limits, diagram (****) is not commutative up to isomorphism, and the notations $u_{!}$ and $u_{! \gamma}$ must be used to avoid confusion.

5.10

Let C be a small category. For every object X of $\widehat{C}$, let C/X denote the category of arrows whose target is X and whose source is an object of C . The source functor defines a functor $j_X : C/X \rightarrow C$. Let $m : Y \rightarrow X$ be a morphism of $\widehat{C}$. Composition of morphisms defines a functor $j_m : C/Y \rightarrow C/X$. The diagram

$$\begin{array}{ccc}
 C/Y & \xrightarrow{j_m} & C/X \\
 & \searrow j_Y & \swarrow j_X \\
 & C &
 \end{array}$$

is commutative. It follows from 5.1 that, for every object F of $\widehat{C/X}$ and every object Y of C , one has

$$j_{X!} F(Y) = \coprod_{u \in \text{Hom}_{\widehat{C}}(Y, X)} F(u). \tag{5.10.1}$$

Formula (5.10.1) allows one to define $j_{X!}$ when C is a $\mathcal{U}$ -category, and one verifies that the functor $j_{X!}$ thus defined is always left adjoint to the functor

$$j_X^* : \widehat{C} \longrightarrow \widehat{C/X}.$$

Proposition 5.11. Let C be a $\mathcal{U}$ -category, and let X be a presheaf on C .

1) The functor

$$j_{X!} : \widehat{C/X} \longrightarrow \widehat{C}$$

factors through the category $\widehat{C}/X$:

$$\widehat{C/X} \xrightarrow{e_X} \widehat{C}/X \longrightarrow \widehat{C}.$$

The functor e_X is an equivalence of categories.

2) The functor

$$e_X \circ j_X^* : \widehat{C} \longrightarrow \widehat{C}/X$$

is canonically isomorphic to the functor

$$H \longmapsto (H \times X \xrightarrow{\text{pr}_2} X).$$

Proof.

1) Let f be the final object of $\widehat{C/X}$. One has a canonical isomorphism

$$f \xrightarrow{\sim} \varinjlim_{Y \in \text{Ob}(C/X)} Y,$$

and hence $j_{X!}(f) = \varinjlim_{Y \in \text{Ob}(C/X)} j_X(Y)$ by 5.4. But $j_{X!}(f) \simeq X$, which gives the factorization. To show that e_X is an equivalence, we merely exhibit a quasi-inverse: to every object $H \rightarrow X$ of $\widehat{C}/X$ associate the presheaf on C/X

$$(Y \rightarrow X) \longmapsto \text{Hom}_{\widehat{C}/X}((Y \rightarrow X), (H \rightarrow X)).$$

2) The functor $e_X \circ j_X^*$ is right adjoint to the forgetful functor, and therefore $e_X \circ j_X^*$ is canonically isomorphic to the functor $H \mapsto (H \times X \xrightarrow{\text{pr}_2} X)$.

5.12

Let $m : Y \rightarrow X$ be a morphism of $\widehat{C}$. By (5.11), the morphism m is canonically isomorphic to the image under e_X of an object of $\widehat{C/X}$, which we shall denote by $[m]$. By restriction to subcategories, the functor e_X defines an equivalence

$$(C/X)/[m] \xrightarrow{e_m} C/Y.$$

The diagram

$$\begin{array}{ccc} (C/X)/[m] & \xrightarrow{e_m} & C/Y \\ & \searrow j_{[m]} & \swarrow j_m \\ & C/X & \end{array}$$

is commutative up to canonical isomorphism.

5.13

We record a result that will be useful in Exposé VI. Let $u : C \rightarrow C'$ be a functor between small categories. For every object H of $\widehat{C}$, denote by

$$u/H : C/H \longrightarrow C'/u_!H$$

the functor which associates to every morphism $m : X \rightarrow H$ the morphism

$$u_!X \xrightarrow{u_!m} u_!H$$

(one knows from 5.4 that one may always set $u_!X = uX$). The following diagram is commutative:

$$\begin{array}{ccc} C/H & \xrightarrow{u/H} & C'/u_!H \\ j_H \downarrow & & \downarrow j_{u_!H} \\ C & \xrightarrow{u} & C'. \end{array} \quad (5.13.1)$$

We therefore have a diagram commutative up to isomorphism; and, since $(u/H)_!$ carries the final object of $\widehat{C/H}$ to the final object of $\widehat{C'/u_!H}$, the diagram

$$\begin{array}{ccc} \widehat{C/H} & \xrightarrow{(u/H)_!} & \widehat{C'/u_!H} \\ e_H \downarrow & & \downarrow e_{u_!H} \\ \widehat{C}/H & \xrightarrow{u_!/H} & \widehat{C'}/u_!H \end{array} \quad (5.13.3)$$

is commutative up to isomorphism.

Proposition 5.14. Let $u : C \rightarrow C'$ be a functor between small categories. Suppose that u has the following property:

(PPF) For every object X of C , the functor

$$u/X : C/X \longrightarrow C'/uX$$

is fully faithful.

Then:

- 1) Let f be the final object of $\widehat{C}$. The functor u factors as

$$C = C/f \xrightarrow{u/f} C'/u_!f \xrightarrow{j_{u_!f}} C'.$$

The functor u/f is fully faithful.

- 2) The functor $u_! : \widehat{C} \rightarrow \widehat{C'}$ factors as

$$\widehat{C} = \widehat{C}/f \xrightarrow{(u/f)_!} \widehat{C'}/u_!f \xrightarrow{e_{u_!f}} \widehat{C'}/u_!f \longrightarrow \widehat{C'},$$

where the functor $(u/f)_!$ is fully faithful, the functor $e_{u_!f}$ is an equivalence, and $\widehat{C'}/u_!f \rightarrow \widehat{C'}$ is the forgetful functor.

- 3) In particular, the functor $u_!$ is faithful and consequently the adjunction morphism $\text{id} \xrightarrow{\Phi} u^*u_!$ is a monomorphism. Moreover, for every morphism $\alpha : H \rightarrow K$ of $\widehat{C}$, the diagram

$$\begin{array}{ccc} H & \xrightarrow{\Phi(H)} & u^*u_!H \\ \alpha \downarrow & & \downarrow u^*u_!(\alpha) \\ K & \xrightarrow{\Phi(K)} & u^*u_!K \end{array}$$

is cartesian.

Proof.

- 1) The factorization comes from diagram (5.13.1). The functor u is faithful. Hence u/f is faithful. Let us show that it is fully faithful. Let X and Y be two objects of C , let $\text{can}_X : uX \rightarrow u_!f$ and $\text{can}_Y : uY \rightarrow u_!f$ be the canonical morphisms, and let

$$\begin{array}{ccc} uX & \xrightarrow{m} & uY \\ & \searrow \text{can}_X & \swarrow \text{can}_Y \\ & u_!f & \end{array}$$

be a morphism of $C'/u_!f$. We have $u_!f = \varinjlim_{Z \in \text{Ob } C} uZ$, hence by 3.1,

$$\text{Hom}_{\widehat{C}'}(uX, u_!f) = \varinjlim_{Z \in \text{Ob } C} \text{Hom}_{C'}(uX, uZ).$$

By the definition of the inductive limit, to say that $\text{can}_Y m = \text{can}_X$ is equivalent to saying that there exist

- a) a finite sequence of objects X_i of C , $i \in [0, n]$, with $X_0 = X$ and $X_n = Y$;
- b) for every i , a morphism $m_i : uX \rightarrow uX_i$, with $m_0 = \text{id}_X$ and $m_n = m$;
- c) for every pair $(i, i+1)$, a morphism $f_i : X_i \rightarrow X_{i+1}$ or a morphism $f_i : X_{i+1} \rightarrow X_i$;

such that the corresponding triangular diagrams

$$\begin{array}{ccc} & uX & \\ m_i \swarrow & & \searrow m_{i+1} \\ uX_i & \xrightarrow{u(f_i)} & uX_{i+1} \end{array} \quad \text{or} \quad \begin{array}{ccc} & uX & \\ m_i \swarrow & & \searrow m_{i+1} \\ uX_i & \xleftarrow{u(f_i)} & uX_{i+1} \end{array}$$

are commutative. One then proves immediately, by induction on i and using property (PPF), that m_i is of the form $u(p_i)$. In particular $m = u(p)$, and hence u/f is fully faithful.

- 2) The factorization is immediate. The functor $(u/f)_!$ is fully faithful by 5.6. The other assertions follow from 5.11.
- 3) The functor $u_!$ is the composite of the forgetful functor, which is faithful, and fully faithful functors. It is therefore faithful. It follows, from the general properties of adjoint functors, that the adjunction morphism $\text{id} \rightarrow u^*u_!$ is a monomorphism. By 2), the functor $u_!$ appears

as the composite of a fully faithful functor $v : \widehat{C} \rightarrow \widehat{C'}/u_!f$ and the forgetful functor. The functor u^* , right adjoint to $u_!$, is therefore the composite of the “product by $u_!f$ ” functor, right adjoint to the forgetful functor, and a functor w right adjoint to v . Moreover, since v is fully faithful, the adjunction morphism $\text{id} \rightarrow wv$ is an isomorphism. The last assertion follows easily.

Exposé I. Presheaves

Sections 6–7

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Translator's status note. This is a working English translation of SGA 4, Exposé I, Sections 6–7. Obvious typographical slips in the working French $\text{\TeX}$ source have been corrected where the mathematics fixes the reading; they are recorded in the accompanying worklog.

6 Faithful functors and conservative functors

Definition 6.1. Let E be a category, and let

$$(\varphi_i : E \rightarrow F_i)_{i \in I}$$

be a family of functors. We say that the family (φ_i) is *faithful* if, for every pair of objects X, Y of E and every pair of arrows $u, v : X \rightrightarrows Y$, the equalities $\varphi_i(u) = \varphi_i(v)$ for all $i \in I$ imply $u = v$. In other words, for every X, Y the map

$$\text{Hom}_E(X, Y) \longrightarrow \prod_i \text{Hom}_{F_i}(\varphi_i(X), \varphi_i(Y))$$

defined by the φ_i is injective. We say that (φ_i) is *conservative* if every arrow u of E such that $\varphi_i(u)$ is an isomorphism for all $i \in I$ is itself an isomorphism. We say that (φ_i) is conservative for monomorphisms, respectively for epimorphisms, etc., if the preceding condition is verified whenever u is a monomorphism, respectively an epimorphism, etc.

6.1.1

If one introduces the single functor

$$\varphi : E \longrightarrow F = \prod_{i \in I} F_i$$

defined by the family (φ_i) , then it is clear that the family is faithful, respectively conservative, respectively conservative for monomorphisms, etc., if and only if φ is faithful, respectively conservative, etc. Here, of course, this means that the one-object family consisting of φ has the corresponding property. Thus there would be no serious loss in subsequently restricting to the case of a family consisting of a single functor. For convenience of later reference, however, we give the following statements for families.

The notions of 6.1 are especially useful when the φ_i satisfy suitable exactness properties; in that case they tend to coincide.

Proposition 6.2. The notation is that of 5.1.

- (i) Suppose that kernels of pairs of arrows, or cokernels of pairs of arrows, are representable in E , and that the φ_i commute with them. Then

$$(\varphi_i) \text{ conservative} \implies (\varphi_i) \text{ faithful.}$$

- (ii) Suppose that fiber products, respectively amalgamated sums, are representable in E , and that the φ_i commute with them. Suppose that (φ_i) is faithful or conservative. Then, for every arrow u of E , the arrow u is a monomorphism, respectively an epimorphism, if and only if, for every $i \in I$, the arrow $\varphi_i(u)$ has the corresponding property.
- (iii) Suppose that in E fiber products and amalgamated sums are representable, that the φ_i commute with them, and that every arrow in E which is a bimorphism, i.e. both a monomorphism and an epimorphism, is an isomorphism. Then

$$(\varphi_i) \text{ faithful} \implies (\varphi_i) \text{ conservative.}$$

- (iv) Suppose that in E fiber products, respectively amalgamated sums, are representable, and that the φ_i commute with them. If (φ_i) is conservative for monomorphisms, respectively for epimorphisms, then (φ_i) is conservative.
- (v) Let D be a diagram type, let $F : d \mapsto F(d)$ be a diagram of type D in E , let X be an object of E , and let $u = (u_d)_{d \in D}$ be a family of arrows

$$X \longrightarrow F(d) \quad (\text{respectively } F(d) \longrightarrow X).$$

Suppose that (φ_i) is conservative, that projective limits, respectively inductive limits, of type D are representable in E , and that the φ_i commute with them. Then u makes X a projective limit, respectively an inductive limit, of F in E if and only if, for every $i \in I$, $\varphi_i(u)$ makes $\varphi_i(X)$ a projective limit, respectively an inductive limit, of $\varphi_i(F)$ in F_i .

Proof. For (i), in the non-dual statement, given a pair of arrows $u, v : X \rightrightarrows Y$, it suffices to express the equality $u = v$ by the condition that the inclusion

$$\text{Ker}(u, v) \longrightarrow X$$

is an isomorphism. Here, and below, we do not repeat the dual argument for the dual statement.

For (ii), if (φ_i) is faithful, one expresses the condition that $u : X \rightarrow Y$ be a monomorphism by the equality $\text{pr}_1 = \text{pr}_2$ for the fiber product $X \times_Y X$. If (φ_i) is conservative, one expresses it by the condition that the diagonal morphism

$$\delta : X \longrightarrow X \times_Y X$$

be an isomorphism.

For (iv), in the latter argument the morphism δ is a monomorphism, so it would actually have sufficed to assume that (φ_i) is conservative for monomorphisms. This then implies that (φ_i) is conservative tout court. Indeed, if $u \in \text{Fl}(E)$ is such that the $\varphi_i(u)$ are isomorphisms, then by what precedes they are monomorphisms, and hence are isomorphisms by the hypothesis on (φ_i) .

Assertion (iii) is a trivial consequence of (ii), and (v) is a trivial consequence of the definitions.

We note the following consequence of (i), (ii), and (iv).

Corollary 6.3. Suppose that in E fiber products and amalgamated sums are representable and that the φ_i commute with them, and that kernels of pairs of arrows or cokernels of pairs of arrows are representable and that the φ_i commute with them. It suffices, for example, that finite projective limits and finite inductive limits be representable in E , and that the φ_i be exact functors. Then the following conditions are equivalent:

- a) (φ_i) is faithful;
- b) (φ_i) is conservative;
- c) (φ_i) is conservative for monomorphisms;
- (c') (φ_i) is conservative for epimorphisms.

We also record the following fact.

Proposition 6.4. Let $\varphi : E \rightarrow F$ be a functor admitting a right adjoint ψ , so that

$$\mathrm{Hom}_F(\varphi(X), Y) \simeq \mathrm{Hom}_E(X, \psi(Y)).$$

For φ , respectively ψ , to be faithful it is necessary and sufficient that, for every object X of E , respectively every object Y of F , the adjunction morphism

$$X \longrightarrow \psi\varphi(X)$$

be a monomorphism. For φ , respectively ψ , to be fully faithful it is necessary and sufficient that the preceding adjunction morphism be an isomorphism.

Indeed, if X', X are two objects of E , the map

$$\mathrm{Hom}_E(X', X) \longrightarrow \mathrm{Hom}_F(\varphi(X'), \varphi(X)) \quad (*)$$

is identified with the map obtained from

$$X \longrightarrow \psi\varphi(X) \quad (**)$$

by applying the functor $\mathrm{Hom}_E(X', -)$. Thus, for $(*)$ to be a monomorphism, respectively an isomorphism, for all X' , with X fixed, it is necessary and sufficient that the adjunction morphism $(**)$ be a monomorphism, respectively an isomorphism.

Proposition 6.5. Let $p : E' \rightarrow E$ be a functor. The following conditions are equivalent:

- (i) p is faithful, conservative, and fibred (*SGA 1, VI, 6.1*).
- (ii) p is a fibred functor whose fibers are discrete categories.
- (iii) For every $X' \in \mathrm{Ob}(E')$, the functor

$$E'_{/X'} \longrightarrow E_{/p(X')}$$

induced by p is an equivalence of categories, surjective on objects.

- (iv) If E is a $\mathcal{U}$ -category, there exists an object $F \in \mathrm{Ob}(\widehat{E})$ and an equivalence of categories over E (*SGA 1, VI, 4.3*)

$$E' \xrightarrow{\sim} E_{/F},$$

where $E_{/F}$ is the full subcategory of $\widehat{E}_{/F}$ consisting of the arrows $X \rightarrow F$ whose source is in E .

6.5.1

Recall that a category C is said to be *discrete* if it is a *groupoid*, i.e. every arrow is invertible, and if it is *rigid*, i.e. the automorphism group of every object is reduced to the unit group. Equivalently, the category is equivalent to the category C' defined by a set I , with $\text{Ob}(C') = I$ and with identity arrows as its only arrows. When C is already assumed to be a groupoid, saying that C is discrete amounts to saying that, for any two objects X, Y of C , there is at most one arrow from X to Y , i.e. that C is isomorphic to the category defined by a preordered set.

The equivalence of conditions (i) and (ii) in 6.5 is an immediate consequence of these reminders and of the following lemma.

Lemma 6.5.2. Let $p : E' \rightarrow E$ be a fibred functor. Then:

- (i) for p to be conservative it is necessary and sufficient that its fiber categories be groupoids;
- (ii) for p to be faithful it is necessary and sufficient that its fiber categories be ordered categories.

Proof. [Proof of 6.5.2] Suppose first that p is conservative. For every arrow u in a fiber E'_X , one has $p(u) = \text{id}_X$, an isomorphism; hence u is an isomorphism in E' , and also in E'_X , since an inverse of u in E' is evidently an inverse in E'_X . Thus E'_X is a groupoid. Conversely, suppose that the E'_X are groupoids, and let u' be an arrow of E' such that $p(u')$ is an isomorphism. We prove that u' is an isomorphism. Since p is fibred, one can factor

$$u' : X' \longrightarrow Y'$$

as a composite

$$X' \longrightarrow u^*(Y') \longrightarrow Y',$$

where the first arrow is an X -morphism, with $X = p(X')$ and $u = p(u')$, and the second is a cartesian morphism over u . The first arrow is an isomorphism because E'_X is a groupoid, and the second is an isomorphism because a cartesian morphism in a fibred category is evidently an isomorphism as soon as its projection is an isomorphism.

Now suppose that p is faithful, and let X', Y' be two objects of a fiber category E'_X . Two arrows from X' to Y' lie over the same arrow id_X of E , hence are identical; therefore E'_X is ordered. Conversely, suppose that the fiber categories are ordered, and prove that p is faithful. Let $u', v' : X' \rightrightarrows Y'$ be arrows of E' over the same arrow $u : X \rightarrow Y$ of E . They factor as

$$X' \rightrightarrows u^*(Y') \longrightarrow Y',$$

where the two arrows $X' \rightrightarrows u^*(Y')$ are arrows of E'_X with the same source and the same target. These are therefore equal, and hence $u' = v'$. $\square$

We return to the proof of 6.5. We have proved (i) $\Leftrightarrow$ (ii). On the other hand, (ii) $\Leftrightarrow$ (iv) is fairly clear: indeed, on one hand the category $E_{/F}$ is fibred over E with fibers the discrete categories defined by the sets $F(X)$, as follows at once from the definitions; on the other hand, if p is as in (ii), then by the sorites of SGA 1, VI, 8, the category E' fibred over E is E -equivalent to the split category over E defined by the functor $F : E^{\text{op}} \rightarrow \mathbf{Set}$ which sends an object X to the set of isomorphism classes of objects of E'_X . This split category is E -isomorphic to $E_{/F}$.

Since (iv) $\Rightarrow$ (iii) is clear, it remains only to prove (iii) $\Rightarrow$ (i). But it is clear that, for p to be faithful, respectively conservative, it is necessary and sufficient that the induced functors

$$E'_{/X'} \longrightarrow E_{/p(X')}$$

be faithful, respectively conservative; *a fortiori*, it suffices that these functors be fully faithful. It remains only to prove that (iii) implies that p is fibred. But again one sees that a functor p is fibred if and only if the induced functors $E'_{/X'} \rightarrow E_{/p(X')}$ are fibred. This is so, in particular, if they are equivalences of categories surjective on objects.

7 Generating and cogenerating subcategories

Definition 7.1. Let E be a category and let C be a full subcategory of E . We say that C is a subcategory of E *generating by strict epimorphisms*, respectively *generating by epimorphisms*, if, for every object X of E , the family of arrows of E with target X and with source $X' \in \text{Ob}(C)$ is strictly epimorphic, respectively epimorphic (1.3). We say that C is a *generating subcategory*, respectively *generating for monomorphisms*, respectively *generating for strict monomorphisms*, if, for every arrow $u : Y \rightarrow X$, respectively every monomorphism of E , respectively every strict monomorphism of E (10.5), such that for every $X' \in \text{Ob}(C)$ the corresponding map

$$\text{Hom}_E(X', Y) \longrightarrow \text{Hom}_E(X', X)$$

is bijective, u is an isomorphism. Finally, we say that a family (X_i) of objects of E generates by strict epimorphisms, respectively etc., if the full subcategory C of E generated by this family generates by strict epimorphisms, respectively etc.

7.1.1

In terms of the family $(h_{X'})_{X' \in \text{Ob}(C)}$ of functors represented by the objects $X' \in \text{Ob}(C)$,

$$h_{X'} : E \longrightarrow \mathbf{Set} \quad (X' \in \text{Ob}(C)),$$

one can express the condition that C be generating, respectively generating for monomorphisms, respectively generating for strict monomorphisms, by saying that the family $(h_{X'})$ is conservative, respectively conservative for monomorphisms, respectively conservative for strict monomorphisms (6.1). It follows immediately from the definitions as well that C generates by epimorphisms if and only if the family $(h_{X'})_{X' \in \text{Ob}(C)}$ is faithful (6.1). We shall also give below, in 7.2(i), an analogous interpretation of the condition that C generate by strict epimorphisms.

7.1.2

As with the notions introduced in 6.1, the notions in 7.1 are especially useful when E has suitable exactness properties; then the various notions introduced have a marked tendency all to be equivalent (7.3). For this reason the question of which of the notions in 7.1 should be considered the most important hardly arises: in the most important cases they coincide, and the term “generating subcategory” may therefore be interpreted indifferently as referring to any of the properties considered in 7.1, for example the first one, which is the strongest of all, as we shall see in 7.2(ii).

7.1.3

Suppose that E is a $\mathcal{U}$ -category, and consider the canonical functor

$$\varphi : E \longrightarrow \widehat{C} = \text{Hom}(C, \mathcal{U}\text{-}\mathbf{Set}), \quad (7.1.3.1)$$

obtained by composing $E \rightarrow \widehat{E} \rightarrow \widehat{C}$, where the first functor is the canonical functor (1.3.3), and the second is the restriction functor to C . It is evident that saying that φ is conservative, respectively faithful, is the same as saying that the family of functors

$$h_{X'} : X \longmapsto \text{Hom}_E(X', X) = \varphi(X)(X')$$

for variable $X' \in \text{Ob}(C)$ is a conservative, respectively faithful, family, i.e. also, by 7.1.2, that C is generating, respectively generating by epimorphisms. Similarly, φ is conservative for monomorphisms, respectively for strict monomorphisms, if and only if the family of the $h_{X'}$ is conservative for monomorphisms, respectively for strict monomorphisms; equivalently, if and only if the subcategory C of E is generating for monomorphisms, respectively for strict monomorphisms.

Proposition 7.2. Let E be a $\mathcal{U}$ -category, and let C be a full subcategory.

(i) The following conditions are equivalent:

- a) C is a subcategory generating by strict epimorphisms.
- b) For every $X \in \text{Ob}(E)$, if $C_{/X}$ denotes the full subcategory of $E_{/X}$ formed by the arrows $X' \rightarrow X$ with $X' \in \text{Ob}(C)$, then the natural arrow from the inclusion functor $j : C_{/X} \rightarrow E$ to the constant functor on $C_{/X}$ with value X makes X an inductive limit of j :

$$X \xleftarrow{\sim} \varinjlim_{C_{/X}} X'.$$

- c) The canonical functor φ of (7.1.3.1) is fully faithful.

(ii) Among the notions of 7.1 one has the following implications:

- 1) C generates by strict epimorphisms (φ fully faithful)
- $\Downarrow \searrow$
- 2) C generates by epimorphisms (φ faithful) 3) C is generating (φ conservative)
- $\Downarrow$
- 4) C generates for monomorphisms (φ conservative for monomorphisms)
- $\Downarrow$
- 5) C generates for strict monomorphisms (φ conservative for strict monomorphisms).

(iii) One has the following conditional implications:

- a) If in E epimorphic families of arrows are strictly epimorphic, then 2) $\Rightarrow$ 1). If in E monomorphisms are strict, then 5) $\Rightarrow$ 4).
- b) If in E kernels of pairs of arrows, respectively fiber products, are representable, then 3) $\Rightarrow$ 2), respectively 4) $\Rightarrow$ 3).

- c) If in E every family of morphisms $X_i \rightarrow X$ with common target X factors as a strictly epimorphic, respectively epimorphic, family $X_i \rightarrow Y$ followed by a monomorphism, respectively by a strict monomorphism, $Y \rightarrow X$, then 4) $\Rightarrow$ 1), respectively 5) $\Rightarrow$ 2).

We immediately record the following corollary.

Corollary 7.3. All the notions considered in 7.1, and recalled in the implication diagram of 7.2(ii), are equivalent in each of the following two cases:

- (i) In E , kernels of pairs of arrows and fiber products are representable, monomorphisms are strict, and epimorphic families of arrows are strictly epimorphic.
- (ii) In E , every family $(X_i \rightarrow X)_{i \in I}$ of arrows with common target X factors as an epimorphic family $(X_i \rightarrow Y)$ followed by a monomorphism $Y \rightarrow X$, every monomorphism of E is strict, and every epimorphic family of arrows of E is strict.

Indeed, in case (i) one has 4) $\Rightarrow$ 3) and 3) $\Rightarrow$ 2) by (b), and 5) $\Rightarrow$ 4) and 2) $\Rightarrow$ 1) by (a). In case (ii) one has, by (c), the implications 4) $\Rightarrow$ 1) and 5) $\Rightarrow$ 2). The conclusion follows from the implication diagram 7.2(ii).

Proof. [Proof of 7.2] For (i), the implication b) $\Rightarrow$ a) follows at once from the definitions. We prove a) $\Rightarrow$ b). Under hypothesis a), one must prove that, for every $X, Y \in \text{Ob}(E)$, every system of arrows

$$u_{X'} : X' \longrightarrow Y$$

indexed by the $X' \in \text{Ob}(C/X)$ such that $u_{X'}f = u_{X''}$ for every arrow $f : X'' \rightarrow X'$ in C/X factors through an arrow, necessarily unique by hypothesis a), $X \rightarrow Y$. By a), it suffices to check that, for every object Z of E/X and every pair of morphisms $v' : Z \rightarrow X'$ and $v'' : Z \rightarrow X''$ in E/X , with X' and X'' in C/X , one has $u_{X'}v' = u_{X''}v''$. But by hypothesis a), the family of arrows $w : X''' \rightarrow Z$, with $X''' \in \text{Ob}(C)$, is epimorphic; hence it suffices to check that for every such w one has

$$(u_{X'}v')w = (u_{X''}v'')w,$$

which is also written $u_{X'}(v'w) = u_{X''}(v''w)$ and follows immediately from the hypothesis on the family u .

We now prove the equivalence of b) and c). For every $X \in \text{Ob}(E)$, the object $\varphi(X)$ of $\widehat{C}$ is the inductive limit in $\widehat{C}$ of the canonical functor $\widehat{C}_{/\varphi(X)} \rightarrow \widehat{C}$ (3.4). But $\widehat{C}_{/\varphi(X)}$ is canonically isomorphic to C/X , so that in $\widehat{C}$ one has

$$\varphi(X) = \varinjlim_{C/X} X',$$

where the object X' of C is identified with the functor in $\widehat{C}$ that it represents. Consequently, for a second object Y of E , there is a canonical isomorphism

$$\begin{aligned} \text{Hom}_{\widehat{C}}(\varphi(X), \varphi(Y)) &\simeq \varinjlim_{C/X} \text{Hom}_{\widehat{C}}(X', \varphi(Y)) \\ &\simeq \varinjlim_{C/X} \varphi(Y)(X') \\ &\simeq \varinjlim_{C/X} \text{Hom}_E(X', Y). \end{aligned} \tag{*}$$

Thus the map

$$\mathrm{Hom}_E(X, Y) \longrightarrow \mathrm{Hom}_{\widehat{C}}(\varphi(X), \varphi(Y))$$

defined by φ is, through the isomorphism between the extreme terms of $(*)$, exactly the map

$$\mathrm{Hom}_E(X, Y) \longrightarrow \varinjlim_{C/X} \mathrm{Hom}_E(X', Y)$$

induced by the inductive system of arrows $X' \rightarrow X$ indexed by C/X considered in 7.2(i)b). Therefore the first map is bijective for every Y , with X fixed, if and only if X is an inductive limit of the inclusion functor $j : C/X \rightarrow E$. This proves the equivalence of b) and c).

For (ii), the implications $1) \Rightarrow 2)$ and $3) \Rightarrow 4) \Rightarrow 5)$ are trivial by the definitions. The implication $1) \Rightarrow 3)$ is obtained by interpreting property 1) as the full faithfulness of φ , by (i), and observing that a fully faithful functor is conservative. We have already observed in 7.1.1 that 3) means that φ is conservative.

For (iii), assertion (a) is a tautology. Assertion (b) follows from 6.2(i), respectively 6.2(iv), applied to the functor φ , noting that φ is left exact. Finally we prove (c). For $X \in \mathrm{Ob}(E)$, consider the family of all morphisms $X_i \rightarrow X$ with $X_i \in \mathrm{Ob}(C)$. By the hypothesis on E , it factors as a strictly epimorphic, respectively epimorphic, family $X_i \rightarrow Y$ followed by a monomorphism, respectively by a strict monomorphism, $Y \rightarrow X$. Then hypothesis 4), respectively 5), implies that $Y \rightarrow X$ is an isomorphism. Thus the family in question is strictly epimorphic, respectively epimorphic. $\square$

Proposition 7.4. Let E be a category, let C be a full subcategory of E , and let X be an object of E . Suppose that C is generating in E , respectively that C generates for monomorphisms and that the fiber product over X of two subobjects of X is representable in E . Then a strict subobject (10.11), respectively a subobject, X' of X is known once, for every $T \in \mathrm{Ob}(C)$, one knows the subset of $\mathrm{Hom}_E(T, X)$ which is the image of $\mathrm{Hom}_E(T, X')$. Consequently, the cardinal of the set of strict subobjects, respectively of the set of subobjects, of X is bounded above by

$$\prod_{T \in \mathrm{Ob}(C)} 2^{\mathrm{card} \mathrm{Hom}_E(T, X)},$$

and if $X \in \mathrm{Ob}(C)$, it is bounded above by $2^{\mathrm{card} \mathrm{Fl}(C)}$.

We first prove the statement with “respectively”. Let X' and X'' be two subobjects of X such that, for every $T \in \mathrm{Ob}(C)$, the images of $\mathrm{Hom}_E(T, X')$ and $\mathrm{Hom}_E(T, X'')$ in $\mathrm{Hom}_E(T, X)$ are equal. They are therefore also equal to the image of $\mathrm{Hom}_E(T, X''')$, where X''' is the fiber product of X' and X'' over X . Since C generates for monomorphisms, it follows that the monomorphisms $X''' \rightarrow X'$ and $X''' \rightarrow X''$ are isomorphisms. Hence X' and X'' are equal, each being equal to the subobject X''' of X .

We now prove the non-respective assertion. By the definition of strict subobject, it suffices to check that knowing the subset $\mathrm{Hom}_E(T, X')$ of $\mathrm{Hom}_E(T, X)$ for every $T \in \mathrm{Ob}(C)$ determines the set of pairs of arrows $u, v : X \rightrightarrows T$ such that $ui = vi$, where $i : X' \rightarrow X$ is the canonical injection. Since C is generating, the relation $ui = vi$ is equivalent to the relation $(ui)f = (vi)f$ for every $f \in \mathrm{Hom}_E(T, X')$, i.e. to $ug = vg$ for every $g \in \mathrm{Hom}_E(T, X)$ coming from $\mathrm{Hom}_E(T, X')$ (that is, of the form if with $f \in \mathrm{Hom}_E(T, X')$). This proves the assertion.

Corollary 7.5. Let E be a category, let C be a full generating subcategory, and let X be an object of C . Then a strict quotient (10.8) X' of X is known once, for every $T \in \mathrm{Ob}(C)$,

one knows the subset of $\text{Hom}_E(T, X)^2$ formed by the pairs (u, v) such that $qu = qv$, where $q : X \rightarrow X'$ is the canonical morphism. Thus the cardinal of the set of strict quotients of X is bounded above by

$$\prod_{T \in \text{Ob}(C)} 2^{\text{card Hom}_E(T, X)^2}.$$

Indeed, by definition, a strict quotient X' of X is known once, for every object Y of E , one knows the subset of $\text{Hom}_E(Y, X) \times \text{Hom}_E(Y, X)$ formed by the pairs (u, v) such that $qu = qv$, where $q : X \rightarrow X'$ is the canonical morphism. But the relation $qu = qv$ is equivalent to $(qu)f = (qv)f$ for every morphism $f : T \rightarrow Y$ whose source T lies in C , since C is generating. This relation may also be written $q(uf) = q(vf)$, which proves the first assertion of 7.5. The second follows immediately.

7.5.1

One can generalize 7.5 by introducing, for a family of objects $(X_i)_{i \in I}$ of E , the notion of a *strict quotient in E of the family*. By this we mean a strictly epimorphic family $(p_i : X_i \rightarrow X')_{i \in I}$ of morphisms of E , with the convention, as for ordinary quotients, that two such families $(p_i : X_i \rightarrow X')$ and $(q_i : X_i \rightarrow X'')$ are identified if there exists an isomorphism $v : X' \rightarrow X''$ (necessarily unique) such that $vp_i = q_i$ for every $i \in I$. With this terminology, the proof of 7.5 applies *mutatis mutandis* to give the following variant.

Variant 7.5.2. Let E and C be as in 7.5, and let $(X_i)_{i \in I}$ be a family of objects of E . Then a strict quotient X' of $(X_i)_{i \in I}$ in E (7.5.1) is known once, for every $T \in \text{Ob}(C)$ and every pair $(i, j) \in I \times I$, one knows the subset of $\text{Hom}_E(T, X_i) \times \text{Hom}_E(T, X_j)$ formed by the pairs (u, v) such that $p_i u = p_j v$, where $p_i : X_i \rightarrow X'$ denotes the canonical morphism for each $i \in I$. Consequently, the cardinal of the set of strict quotients of $(X_i)_{i \in I}$ in E is bounded above by

$$\prod_{T \in \text{Ob}(C)} \prod_{i, j \in I} 2^{\text{card Hom}_E(T, X_i) \times \text{card Hom}_E(T, X_j)}.$$

7.5.3

One sees at once that the analogous conclusion is true if one assumes only that C generates for strict monomorphisms, provided one assumes that the products $X_i \times X_j$ are representable and restricts to *effective* quotients of the family $(X_i)_{i \in I}$, i.e. to strict quotients X' such that the fiber products $X_i \times_{X'} X_j$ are representable in E ; this is no restriction if E is stable under fiber products.

Proposition 7.6. Let E be a category, let C be a full subcategory of E generating by epimorphisms (7.1), let Π_0 be an infinite cardinal $\geq \text{card Fl}(C)$, let $\Pi \geq \Pi_0$ be a cardinal, and let $(u_i : X_i \rightarrow X)_{i \in I}$ be a universal epimorphic family (10.3) in E formed by squareable morphisms (10.7), with sources $X_i \in \text{Ob}(C)$, and such that $\text{card}(I) \leq \Pi$. Then

$$\text{card Fl}(C/X) \leq \Pi^{\Pi_0}.$$

Let $T \in \text{Ob}(C)$. It will be enough to prove that

$$\text{card Hom}_E(T, X) \leq \Pi^{\Pi_0}. \tag{7.6.1}$$

for then one successively obtains

$$\text{card Ob}(C_{/X}) \leq (\Pi^{\Pi_0}) \times \text{card Ob}(C) \leq (\Pi^{\Pi_0}) \times \Pi_0 = \Pi^{\Pi_0},$$

and finally

$$\text{card Fl}(C_{/X}) \leq ((\Pi^{\Pi_0})^2) \times \Pi_0 = \Pi^{\Pi_0},$$

since for two objects S, T of $C_{/X}$ one has

$$\text{card Hom}_{C_{/X}}(S, T) \leq \text{card Hom}_C(S, T) \leq \Pi_0.$$

To prove (7.6.1), first note the following lemma.

Lemma 7.6.2. Let J be a set such that $\text{card}(J) = \Pi_0$. For every object T of C and every morphism $f : T \rightarrow X$, there exists an epimorphic family $(v_j : S_j \rightarrow T)_{j \in J}$, with sources $S_j \in \text{Ob}(C)$, and for every $j \in J$ an element $i(j) \in I$ and a morphism $g_j : S_j \rightarrow X_{i(j)}$, such that

$$u_{i(j)} g_j = f v_j.$$

Indeed, the family of projections

$$X_i \times_X T \longrightarrow T$$

is epimorphic by hypothesis. On the other hand, since C generates by epimorphisms, for every i the family of arrows $S \rightarrow X_i \times_X T$ with source $S \in \text{Ob}(C)$ is epimorphic. Thus, by transitivity, the family of arrows $v : S \rightarrow T$ with source in C which factor through one of the $X_i \times_X T$ is epimorphic. These are precisely the arrows $v : S \rightarrow T$ with source in C for which there exist $i \in I$ and $g : S \rightarrow X_i$ such that $u_i g = f v$. Since the set of these arrows $v : S \rightarrow T$ is contained in $\text{Fl}(C)$ and therefore has cardinal $\leq \Pi_0$, it can be indexed by the index set J . This proves the lemma.

Now a morphism $f : T \rightarrow X$ is known once the $f v_j$ are known, and these are known once the g_j are known. Thus $\text{card Hom}_E(T, X)$ is bounded above by the cardinal of the set of families $(i(j), S_j, v_j, g_j)_{j \in J}$. The cardinal of the set of maps $j \mapsto i(j)$ from J to I is $\leq \Pi^{\Pi_0}$; and for such a map fixed, the cardinal of the set of the corresponding families S_j, g_j, v_j is bounded above by the cardinal of the set of maps from J to $\text{Fl}(C) \times \text{Fl}(C)$, which is $\leq \Pi_0^{\Pi_0}$, since $\text{card}(\text{Fl}(C) \times \text{Fl}(C)) = \Pi_0^2 = \Pi_0$. Thus the left hand side of (7.6.1) is bounded above by

$$\Pi^{\Pi_0} \times \Pi_0^{\Pi_0} = \Pi^{\Pi_0},$$

which completes the proof of 7.6.

Proposition 7.7. Let E be a category in which fiber products are representable, let $(X_\alpha)_{\alpha \in A}$ be a generating family of objects of E , and let $(\varphi_i)_{i \in I}$ be a family of functors $\varphi_i : E \rightarrow E_i$ commuting with fiber products. For (φ_i) to be conservative (6.1), it is necessary and sufficient that for every $\alpha \in A$ and every subobject X' of X_α distinct from X_α , there exist $i \in I$ such that the morphism

$$\varphi_i(X') \longrightarrow \varphi_i(X_\alpha)$$

is not an isomorphism. In this case, if E is a $\mathcal{U}$ -category and A is $\mathcal{U}$ -small, there exists a $\mathcal{U}$ -small subset J of I such that $(\varphi_j)_{j \in J}$ is already a conservative family of functors.

The necessity of the condition of conservativity under consideration is evident; we prove its sufficiency. By 6.2(iv), it suffices to prove that (φ_i) is conservative for monomorphisms. Let $u : Y' \rightarrow Y$ be a monomorphism in E which is not an isomorphism. We must prove that there is an $i \in I$ such that $\varphi_i(u)$ is not an isomorphism. By the hypothesis on (X_α) , there exists an α and a morphism $X_\alpha \rightarrow Y$ which does not factor through Y' . In other words, the inverse image X' of Y' is a subobject of X_α distinct from X_α . By hypothesis, there exists an $i \in I$ such that

$$\varphi_i(X') \longrightarrow \varphi_i(X_\alpha)$$

is not an isomorphism. Since

$$\varphi_i(X') \simeq \varphi_i(X_\alpha) \times_{\varphi_i(Y)} \varphi_i(Y'),$$

it follows that $\varphi_i(Y') \rightarrow \varphi_i(Y)$ is not an isomorphism.

The second assertion of 7.7 follows at once from the first, taking 7.4 into account.

Corollary 7.7.1. Let E be a $\mathcal{U}$ -category in which fiber products are representable, and suppose that E admits a generating family of objects which is $\mathcal{U}$ -small. Then, for every generating family $(Y_i)_{i \in I}$ of E , there exists a $\mathcal{U}$ -small generating subfamily $(Y_j)_{j \in J}$, with $\text{card}(J) \in \mathcal{U}$.

It suffices to apply 7.7 to a $\mathcal{U}$ -small generating family $(X_\alpha)_{\alpha \in A}$ of E and to the family of functors φ_i represented by the Y_i .

Proposition 7.8. Let E be a category, let C be a full subcategory generating by strict epimorphisms, let D be a category, and let $\text{Hom}'(E, D)$ be the full subcategory of $\text{Hom}(E, D)$ consisting of the functors which commute with inductive limits of type C/X , where X ranges over the objects of E . Then the functor $F \mapsto F|_C$ induces a fully faithful functor

$$\text{Hom}'(E, D) \longrightarrow \text{Hom}(C, D).$$

Consequently, if $\text{Hom}(C, D)$ is a $\mathcal{U}$ -category (1.1), for example by (1.1.1 b)) if C is $\mathcal{U}$ -small and D is a $\mathcal{U}$ -category, then $\text{Hom}'(E, D)$ is also a $\mathcal{U}$ -category.

Let $F, G : E \rightarrow D$ be two functors in the source category, and let $u : F \rightarrow G$ be a morphism. If $X = \varinjlim X_i$ in E and if F and G commute with the inductive limit in question, then $u(X) : F(X) \rightarrow G(X)$ is identified with the limit of the morphisms $u(X_i) : F(X_i) \rightarrow G(X_i)$, and hence is known once the $u(X_i)$ are known. This shows that the functor considered in 7.8 is faithful, by the implication a) $\Rightarrow$ b) in 7.2(i). Conversely, let $v : F|_C \rightarrow G|_C$ be a morphism; we prove that it comes from a morphism $u : F \rightarrow G$. For every $X \in \text{Ob}(E)$ define

$$u(X) : F(X) = \varinjlim_{C/X} F(X_i) \longrightarrow G(X) = \varinjlim_{C/X} G(X_i)$$

as the inductive limit of the $v(X_i)$. It is immediate that this gives a morphism functorial in X , hence a morphism $u : F \rightarrow G$, and finally that the morphism induced by u from $F|_C$ to $G|_C$ is v . This completes the proof.

7.1 Families and subcategories cogenerating

Let E be a category, and let C be a full subcategory of E . We say that C is cogenerating by strict monomorphisms, respectively cogenerating by monomorphisms, respectively cogenerating,

respectively cogenerating for epimorphisms, respectively cogenerating for strict epimorphisms, if the full subcategory C^{op} of E^{op} generates by strict epimorphisms, respectively etc. The terminology for families is analogous. Thus all the results in this section concerning the notion of generating subcategory and its variants (7.1) give corresponding results for the dual notions, which we leave to the reader to formulate for personal satisfaction.

Proposition 7.10. Let E be a category, let C be a full generating subcategory (7.1), and let D be a full subcategory of E . For D to be cogenerating (7.9), it is necessary and sufficient that, for every pair of arrows

$$T \begin{smallmatrix} \xrightarrow{u} \\ \xrightarrow{v} \end{smallmatrix} X$$

in E with source $T \in \text{Ob}(C)$ and with $u \neq v$, there exist an arrow $w : X \rightarrow I$ with target $I \in \text{Ob}(D)$ such that $wu \neq wv$.

By definition, to say that D is cogenerating means that for every pair of arrows $Y \rightrightarrows X$ in E such that $u \neq v$, there exists an arrow $w : X \rightarrow I$, with $I \in \text{Ob}(D)$, such that $wu \neq wv$. Thus 7.10 simply says that it suffices to test this property when $Y \in \text{Ob}(C)$. Since C is generating, the hypothesis $u \neq v$ implies that there exists $f : T \rightarrow Y$ such that $uf \neq vf$. By the hypothesis there exists $w : X \rightarrow I$, with target $I \in \text{Ob}(D)$, such that $w(uf) \neq w(vf)$, and hence $wu \neq wv$. $\square$

Corollary 7.11. With the notation of 7.10, suppose that the objects I of D are injective objects of E , i.e. that for every monomorphism $X \rightarrow Y$ in E , the corresponding map

$$\text{Hom}_E(Y, I) \longrightarrow \text{Hom}_E(X, I)$$

is surjective. Suppose moreover that every pair of arrows

$$T \begin{smallmatrix} \xrightarrow{u} \\ \xrightarrow{v} \end{smallmatrix} X$$

in E factors as an epimorphic, respectively effective epimorphic, pair $T \rightrightarrows X'$ followed by a monomorphism $i : X' \rightarrow X$. Then in the criterion 7.10 for D to be cogenerating, one may restrict to pairs (u, v) which are epimorphic, respectively effective epimorphic.

We conclude:

Corollary 7.12. Let E be a $\mathcal{U}$ -category admitting a small generating subcategory C , and such that every object of E is the source of a monomorphism into an injective object of E . Suppose moreover that, for every $T \in \text{Ob}(C)$, the sum $T \amalg T$ in E is representable, and that every morphism with source $T \amalg T$ factors as an effective epimorphism followed by a monomorphism. Then E admits a small full cogenerating subcategory D . More precisely, one may take D such that

$$\text{card Ob}(D) \leq \prod_{T, T' \in \text{Ob}(C)} 2^{\text{card}(\text{Hom}_E(T', T \amalg T))^2}.$$

Indeed, by 7.11 it is enough, for every $T \in \text{Ob}(C)$ and every effective quotient X of $T \amalg T$, to choose an embedding of X into an injective object I of E , and to take for D the full subcategory of E generated by these I . The conclusion then follows from 7.5.

Examples 7.13. To construct small cogenerating subcategories in terms of small generating subcategories, one is therefore led to look for conditions ensuring that a $\mathcal{U}$ -category E has

“enough injectives”, i.e. that every object embeds into an injective object by a monomorphism. It is well known [Tohoku] that this condition is satisfied in a $\mathcal{U}$ -abelian category with exact small filtered inductive limits (axiom AB5 of *loc. cit.*) admitting a small generating family. The construction of *loc. cit.* is not, moreover, essentially tied to abelian categories, and also works in the category of sheaves of $\mathcal{U}$ -sets on a topological space $X \in \mathcal{U}$. We shall not state here the exactness properties which make the construction in question work, and shall limit ourselves to observing that in the special case of the category of sheaves of sets on X , one immediately reduces to the case of *loc. cit.* as follows.

Note that if $\mathcal{O}_X$ is a ring on X , then every injective object I of the category of $\mathcal{O}_X$ -modules is also injective as an object of the category of sheaves of sets. Indeed, if F is a sheaf of sets and $\mathcal{O}_X[F]$ is the “free $\mathcal{O}_X$ -module generated by F ”, then by definition there is a homomorphism of sheaves of sets

$$F \longrightarrow \mathcal{O}_X[F] \tag{*}$$

giving an isomorphism, functorial in F ,

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X[F], I) \xrightarrow{\sim} \mathrm{Hom}(F, I).$$

Since the functor $F \mapsto \mathcal{O}_X[F]$ plainly carries monomorphisms to monomorphisms, it follows at once that, if I is an injective $\mathcal{O}_X$ -module, then it is also an injective sheaf of sets. It follows that if the morphism $(*)$ is a monomorphism, as happens if one takes for $\mathcal{O}_X$ a constant ring with value a ring $A \neq 0$, then an embedding of $\mathcal{O}_X[F]$ into an injective $\mathcal{O}_X$ -module I gives an embedding of F into the underlying injective sheaf of sets of I .

The same argument applies, without change, to the category of sheaves of $\mathcal{U}$ -sets on a $\mathcal{U}$ -site, which will be introduced in the next exposé.

The interest of the existence of a small cogenerating subcategory lies chiefly in the representability criterion 8.12.7 below.

Exposé I. Presheaves

Section 8.1–8.6

A. Grothendieck and J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé I, Section 8.1 through Section 8.6. The source has one visibly truncated sentence in the proof of Proposition 8.6.4; the translation marks the mathematically forced completion as referring to the immediately following Corollary 8.6.5.

8 Ind-objects and pro-objects

8.1 Cofinal functors and cofinal subcategories

Definition 8.1.1. Let

$$\varphi : I \longrightarrow I' \tag{8.1.1.1}$$

be a functor. We say that φ is a *cofinal functor* if, for every category C and every functor $u : I' \rightarrow C$, considering $\varinjlim u$ and $\varinjlim u \circ \varphi$ as objects of $\text{Hom}(G(\mathcal{V}\text{-}\mathbf{Set}))^{\text{op}}$ (2.1, 2.3.1), where $\mathcal{V}$ is a universe such that $I, I' \in \mathcal{V}$ and C is a $\mathcal{V}$ -category, the canonical morphism

$$\varinjlim u \circ \varphi \longrightarrow \varinjlim u \tag{8.1.1.2}$$

is an isomorphism. When φ is the inclusion functor of a subcategory of I' , we say that I is a *cofinal subcategory* if φ is cofinal.

8.1.2

Notice that, for fixed φ and u , the bijectivity of (8.1.1.2) does not depend on the choice of the universe $\mathcal{V}$.

Returning to the definition of the terms appearing in (8.1.1.2), one sees that to say that φ is cofinal is also to say that, for every universe $\mathcal{V}$ such that I and I' are $\mathcal{V}$ -small, and for every functor

$$v : I'^{\text{op}} \longrightarrow (\mathcal{V}\text{-}\mathbf{Set}),$$

the canonical homomorphism

$$\varprojlim v \longrightarrow \varprojlim v \circ \varphi \tag{8.1.2.1}$$

is an isomorphism, that is, a bijection. To see that this condition is indeed necessary, observe, putting $v = v^{\text{op}}$, that it also says that the condition of Definition 8.1.1 is fulfilled when one takes $C = (\mathcal{V}\text{-}\mathbf{Set})^{\text{op}}$; this implies that the inductive limits considered in (8.1.1.2) are representable in C .

8.1.2.2

It is immediate from the definition that *the composite of two cofinal functors is cofinal*.

Proposition 8.1.3. Let $\varphi : I \rightarrow I'$ be a functor.

- a) For φ to be cofinal, it is necessary that φ satisfy the condition:

F 1) For every object i' of I' , there exists an object i of I such that $\varphi(i)$ dominates i' , i.e. such that $\text{Hom}(i', \varphi(i)) \neq \emptyset$.

b) Suppose that I is filtered. For φ to be cofinal, it is necessary and sufficient that φ satisfy condition (F 1) and the following condition:

F 2) For every object i of I and every double arrow

$$i' \begin{array}{c} \xrightarrow{f'} \\ \xrightarrow{g'} \end{array} \varphi(i)$$

in I' with target $\varphi(i)$, there exists an arrow $h : i \rightarrow j$ in I such that $\varphi(h)f' = \varphi(h)g'$.

Moreover, if φ is cofinal, then I' is filtered.

c) Suppose that I' is filtered and that φ is fully faithful. For φ to be cofinal, it is necessary and sufficient that it satisfy condition F 1) of a). This implies that I is filtered.

Proof. For necessity in a) and b), one uses Definition 8.1.1 only in the case where u is the canonical inclusion functor

$$u : I' \hookrightarrow \widehat{I'}$$

(choosing a universe $\mathcal{U}$ such that I and I' are $\mathcal{U}$ -small). Then (8.1.1.2) may be interpreted as an arrow of $\widehat{I'}$, whose target is the *final* functor on I' . One sees immediately that condition F 1) expresses that this arrow is an epimorphism of $\widehat{I'}$, i.e. that it is surjective on each argument, since inductive limits in $\widehat{I'}$ are calculated argument by argument. This proves a).

Now suppose that I is filtered and that φ is cofinal. Then I' is filtered: indeed, the condition that two objects of I' be dominated by a third follows at once from the corresponding condition on I and from F 1). It remains only to prove condition PS 2) of 2.7, i.e. that for every double arrow

$$f', g' : i' \rightrightarrows j'$$

in I' , there exists an arrow $h' : j' \rightarrow k'$ in I' such that $h'f' = h'g'$. By F 1) we may suppose that k' is of the form $\varphi(i)$. But by the hypothesis that φ is cofinal, for every object i' of I' the set $\varinjlim_i \text{Hom}(i', \varphi(i))$ has one element. It follows from the standard description of filtered inductive limits in **Set** (2.8.1) that there is an arrow $h : i \rightarrow j$ in I such that $\varphi(h)f' = \varphi(h)g'$. This finishes the proof that I' is filtered, and proves F 2) at the same time.

To prove that the conditions stated in b) are sufficient for φ to be cofinal, use the form 8.1.2 of the definition. One immediately checks that F 1) implies that (8.1.2.1) is a monomorphism, i.e. injective on each argument, while F 2), together with F 1), ensures that it is bijective, since projective limits in $\widehat{I'^{\text{op}}}$ are calculated argument by argument. This proves b).

Finally, if I' is filtered and φ is fully faithful, it is immediate that F 1) implies that I is filtered and implies F 2). Thus c) follows from a) and b).

8.1.4

When I' is a filtered category and I is a full subcategory of I' , Proposition 8.1.3 c) shows that the condition that I be cofinal in I' depends only on the subset $\text{Ob } I$ of the preordered set $\text{Ob } I'$ (for the preorder relation “ $x \leq y$ ” iff “ $\text{Hom}(x, y) \neq \emptyset$ ”). If J' is a preordered set and J is a

subset of J' , we shall sometimes say that J is a *cofinal subset* of J' when every element of J' is dominated by an element of J . When J' is filtered, this therefore means that the inclusion functor $\mathcal{J} \hookrightarrow \mathcal{J}'$ of the associated categories is cofinal.

8.1.5

In what follows, we shall use the notion of cofinal functor only in cases where the categories I and I' are filtered. Classically, one even restricted oneself to categories associated with preordered sets, i.e. categories in which there is at most one arrow with given source and target. It turns out, however, that this restriction is inconvenient in applications, because the “natural” filtered categories which arise in many applications are *not* ordered categories. The following result, due to P. Deligne, nevertheless shows that there is no essential difference between the two points of view.

Proposition 8.1.6. Let I be a small filtered category. Then there exists a small ordered set E , and a cofinal functor

$$\varphi : \mathcal{E} \longrightarrow I,$$

where $\mathcal{E}$ denotes the category associated with E .

First suppose that the preordered set $\text{Ob } I$ has no greatest element. By a subdiagram of I we mean a pair $D = (\mathcal{O}, F)$ consisting of a subset F of $\text{Fl}(I)$ and a subset $\mathcal{O}$ of $\text{Ob } I$, such that for every $f \in F$, the source and target of f belong to $\mathcal{O}$. An element e of $\mathcal{O}$ is called a *final object* of the subdiagram D if, for every $x \in D$, the set $\text{Hom}(x, e) \cap F$ of arrows of D with source x and target e has exactly one element f_x , if for every arrow $g : x \rightarrow y$ of D one has $f_x = f_y \circ g$, and if $f_e = \text{id}_e$.

Let E be the set of *finite* subdiagrams D of I having a *unique* final element $\varphi(D)$, and order E by inclusion. If D, D' are elements of E and $D \subset D'$, then there is a unique arrow $\varphi(D', D)$ of D' with source $\varphi(D)$ and target $\varphi(D')$; if $D \subset D' \subset D''$ are inclusions in E , then of course

$$\varphi(D'', D) = \varphi(D'', D')\varphi(D', D),$$

and also $\varphi(D, D) = \text{id}_{\varphi(D)}$. Thus one obtains a functor $\varphi : \mathcal{E} \rightarrow I$, and it remains to prove that E is filtered and that φ is cofinal. By Proposition 8.1.3 b), three verifications are required.

- 1) Condition F 1) of 8.1.3. For every $i \in I$, take D to be the subdiagram of I reduced to the object i and its identity arrow. Then $i \leq \varphi(D) = i$. This condition F 1) also shows that $E \neq \emptyset$.
- 2) Condition F 2). For every $i \in \text{Ob } I$, every $D \in E$, and every double arrow $i \rightrightarrows \varphi(D)$, one must find a diagram $D' \in E$, containing D , such that

$$\varphi(D', D) : \varphi(D) \longrightarrow \varphi(D')$$

equalizes the double arrow. Since I is filtered, there exists an object j of I and an arrow $f : \varphi(D) \rightarrow j$ equalizing the double arrow. Since I has no greatest element, we may suppose that j is a *strict* upper bound of $\varphi(D)$, which implies that it is distinct from all the other objects of D . Let D' be the subdiagram of I whose set of objects is $\mathcal{O} \cup \{j\}$, and whose set of arrows consists of the arrows of D , together with the composites

$$x \xrightarrow{f_x} \varphi(D) \xrightarrow{f} j,$$

where x is an object of D and f_x is the unique arrow of D with source x and target $\varphi(D)$, and together with id_j . It is then clear that D' is a finite subdiagram of I , that it has j as its unique final object, hence that $D' \in E$, and that D' satisfies the required condition.

- 3) Let D and D' be two subdiagrams in E . They are contained in some common $D'' \in E$. Indeed, one can find a strict upper bound j of $\varphi(D)$ and $\varphi(D')$,

$$\begin{array}{ccc} \varphi(D) & & \\ & \searrow f & \\ & & j \\ & \nearrow f' & \\ \varphi(D') & & \end{array}$$

and take D'' to be the subdiagram whose set of objects is the union of the sets of objects of D and D' and of $\{j\}$, and whose set of arrows is the union of the sets of arrows of D and D' , the composites $f \circ f_x$ with x an object of D , the composites $f' \circ f'_{x'}$ with x' an object of D' , and $\{\text{id}_j\}$. This is indeed a finite subdiagram of I . We show that, after replacing j, f, f' by j', gf, gf' with a suitable $g : j \rightarrow j'$, we may arrange for j to be a final object of D'' . This is the same as saying that, for every x which is an object of both D and D' , one has

$$ff_x = f'f'_{x'}.$$

But this is only a *finite* set of double arrows to be equalized by some $g : j \rightarrow j'$, and this is possible because I is filtered.

This completes the proof in the case considered. The general case is reduced to it by introducing the filtered category $\mathbb{N}$ associated with the ordered set $\mathbb{N}$ of natural numbers, and observing that the category $\mathbb{N} \times I$ is filtered, has no greatest element, and that the projection $\mathbb{N} \times I \rightarrow I$ is a cofinal functor.

Corollary 8.1.7. Let I be a $\mathcal{U}$ -category. For there to exist a small filtered ordered set E and a cofinal functor $\mathcal{E} \rightarrow I$, it is necessary and sufficient that I be filtered and that $\text{Ob } I$ admit a small cofinal subset I' (8.1.4).

This is necessary by 8.1.3 a), and sufficient by 8.1.6 applied to the full subcategory of I defined by I' .

Definition 8.1.8. Let I be a filtered category. We say that I is *essentially small* if I is a $\mathcal{U}$ -category and if it satisfies the equivalent conditions of 8.1.7.

8.2 Ind-objects and ind-representable functors

8.2.1

In what follows we fix a $\mathcal{U}$ -category C , which we always regard as embedded in the category $\widehat{C}$ of $\mathcal{U}$ -presheaves by means of the canonical functor (1.3.1)

$$h : C \hookrightarrow \widehat{C}. \quad (8.2.1.1)$$

We shall call an *ind-object* of C (or an *inductive system* of C) any functor

$$\varphi : I \longrightarrow C, \quad (8.2.1.2)$$

where I is a *filtered* category (2.7), called the *index category* of the ind-object. Apart from this, I is arbitrary; in particular we do not require I to be a $\mathcal{U}$ -category. It will often be convenient to denote an ind-object φ by the indexed notation $(X_i)_{i \in \text{Ob } I}$, or even, by a further abuse of notation, by $(X_i)_{i \in I}$, $(X_i)_i$, or (X_i) , where $X_i = \varphi(i)$ for $i \in \text{Ob } I$. This notation is no more and no less abusive than the classical notation for inductive systems indexed by filtered ordered sets, where the transition morphisms are likewise not mentioned. The X_i are called the *component objects* of the ind-object $\mathcal{X}$. Notice that, in the general case considered here, the “*transition morphisms*” $\varphi(f)$ of the inductive system are indexed by the set $\text{Fl}(I)$ of arrows of I , which in general is not identified with a subset of $\text{Ob } I \times \text{Ob } I$. In other words, for given i and j , with j dominating i , there may exist more than one transition morphism from X_i to X_j .

8.2.2

The most useful ind-objects $\mathcal{X} = (X_i)_{i \in I}$ of C are those for which the index category I is essentially small (8.1.8); such an ind-object is called *essentially small*. If $(X_i)_{i \in I}$ is such an object, using the fact that small inductive limits in $\widehat{C}$ are representable (3.1), one may consider

$$L(\mathcal{X}) := \varinjlim h \cdot \mathcal{X} = \varinjlim_i X_i \in \text{Ob } \widehat{C}, \quad (8.2.2.1)$$

where the last limit is taken in $\widehat{C}$. Thus $L(\mathcal{X})$ is the presheaf

$$L(\mathcal{X}) : Y \longmapsto \varinjlim_i \text{Hom}(Y, X_i) \quad (8.2.2.2)$$

on C . We say that this presheaf is the *presheaf ind-represented by the ind-object* $\mathcal{X} = (X_i)_{i \in I}$ of C . A presheaf F on C is called an *ind-representable presheaf* if it is isomorphic to a presheaf $L(\mathcal{X})$ ind-represented by an essentially small ind-object. It follows from 8.1.6 that, in this definition, one may suppose that I is a small category, and even that I is associated with an ordered set belonging to $\mathcal{U}$.

8.2.3

Consider an ind-object $\mathcal{X} = (X_i)_{i \in I}$, given by a functor $\varphi : I \rightarrow C$, and let

$$u : I' \longrightarrow I$$

be a functor, where I' is a second filtered category. Then the functor $\varphi \circ u$ is an ind-object $\mathcal{X}'$ of C with index category I' , written in indexed notation as $(X_{u(i')})_{i' \in I'}$. It is called the ind-object of C *deduced from* $(X_i)_{i \in I}$ *by change of index categories* using the functor u . If I and I' , equivalently $\mathcal{X}$ and $\mathcal{X}'$, are essentially small, one obtains a canonical morphism

$$L(\mathcal{X}') = \varinjlim_{i'} X_{u(i')} \longrightarrow L(\mathcal{X}) = \varinjlim_i X_i \quad (8.2.3.1)$$

between the presheaves ind-represented by $\mathcal{X}'$ and $\mathcal{X}$. When u is a cofinal functor (8.1.1), $\mathcal{X}$ is essentially small if and only if $\mathcal{X}'$ is so (8.1.7), and the preceding homomorphism (8.2.3.1) is an isomorphism

$$L(\mathcal{X}') \simeq L(\mathcal{X}). \quad (8.2.3.2)$$

8.2.4

Let

$$\mathcal{X} = (X_i)_{i \in I}, \quad \mathcal{Y} = (Y_j)_{j \in J} \quad (8.2.4.1)$$

be two essentially small ind-objects of C , indexed by filtered essentially small categories I and J . A *morphism* from the ind-object $\mathcal{X}$ to the ind-object $\mathcal{Y}$ is by definition a morphism

$$L(\mathcal{X}) \longrightarrow L(\mathcal{Y}) \quad (8.2.4.2)$$

between the presheaves they ind-represent. We put

$$\mathrm{Hom}_{\mathrm{ind-ob}}(\mathcal{X}, \mathcal{Y}) = \mathrm{Hom}_{\widehat{C}}(L(\mathcal{X}), L(\mathcal{Y})). \quad (8.2.4.3)$$

We suppress the subscript “ind-ob” on Hom if no confusion is likely. Composition of morphisms of ind-objects of C is defined as composition of morphisms of the presheaves they ind-represent. If $\mathcal{V} \supset \mathcal{U}$ is a universe containing $\mathcal{U}$, and if one restricts to essentially small ind-objects which belong to $\mathcal{V}$, these form the set of objects of a category whose arrows are triples $(\mathcal{X}, \mathcal{Y}, u)$, where $\mathcal{X}$ and $\mathcal{Y}$ are essentially small ind-objects belonging to $\mathcal{V}$ and $u : \mathcal{X} \rightarrow \mathcal{Y}$ is a morphism of ind-objects, i.e. a morphism $L(\mathcal{X}) \rightarrow L(\mathcal{Y})$. The category so obtained is denoted

$$\mathrm{Ind}_{\mathcal{V}}(C, \mathcal{U}). \quad (8.2.4.4)$$

When $\mathcal{V} = \mathcal{U}$, it is denoted simply

$$\mathrm{Ind}_{\mathcal{U}}(C) \quad \text{or} \quad \mathrm{Ind}(C), \quad (8.2.4.5)$$

and is called the *category of ind-objects of C relative to $\mathcal{U}$* , the mention of $\mathcal{U}$ being suppressed when no confusion is likely.

Of course, if $\mathcal{V} \subset \mathcal{V}'$ are two universes containing $\mathcal{U}$, then $\mathrm{Ind}_{\mathcal{V}}(C, \mathcal{U})$ is a full subcategory of $\mathrm{Ind}_{\mathcal{V}'}(C, \mathcal{U})$. It follows from 8.2.3 that the inclusion functor is an *equivalence of categories*

$$\mathrm{Ind}_{\mathcal{V}}(C, \mathcal{U}) \xrightarrow{\sim} \mathrm{Ind}_{\mathcal{V}'}(C, \mathcal{U}). \quad (8.2.4.6)$$

This shows in particular that *every essentially small ind-object of C defines an element of $\mathrm{Ind}(C)$, uniquely up to isomorphism*. This observation justifies the fairly common practical abuse of language which consists in identifying any essentially small ind-object of C with an object of $\mathrm{Ind}(C)$.

It follows at once from the definitions that for every universe $\mathcal{V}$ we have a canonical functor

$$L : \mathrm{Ind}_{\mathcal{V}}(C, \mathcal{U}) \longrightarrow \widehat{C} \quad (8.2.4.7)$$

which is fully faithful. These functors are of course known up to unique isomorphism, by (8.2.4.6), once one of them is known, in particular once one knows the canonical functor

$$L : \mathrm{Ind}(C) \longrightarrow \widehat{C}. \quad (8.2.4.8)$$

Since this functor is fully faithful, it is often used to identify an object of the first category with the functor it pro-represents, or even to identify $\mathrm{Ind}(C)$ with its essential image in $\widehat{C}$, consisting of the ind-representable presheaves. Notice, however, that this identification has distinctly more drawbacks than the analogous identification of C with a full subcategory of $\widehat{C}$ via the functor h (8.2.1.1), because unlike the latter functor, the functor (8.2.4.8) is not in general injective on objects.

8.2.5

Let us spell out definition (8.2.4.3) in terms of the indexed expressions (8.2.4.1) of the ind-objects under consideration. Taking account of the definition (8.2.2.1), one obtains

$$\mathrm{Hom}(\mathcal{X}, \mathcal{Y}) \simeq \varprojlim_i \mathrm{Hom}(X_i, \mathcal{Y}) := \varprojlim_i \mathrm{Hom}(X_i, L(\mathcal{Y})) = \varprojlim_i \varinjlim_j \mathrm{Hom}_C(X_i, Y_j),$$

the last equality coming from (8.2.2.2) applied to $\mathcal{Y}$. Thus there is a canonical bijection

$$\mathrm{Hom}((X_i)_{i \in I}, (Y_j)_{j \in J}) \simeq \varprojlim_{i \in I} \varinjlim_{j \in J} \mathrm{Hom}_C(X_i, Y_j). \quad (8.2.5.1)$$

We leave to the reader the task of making the composition of morphisms of ind-objects explicit using this formula. Notice that it follows at once from this formula that the set $\mathrm{Hom}(\mathcal{X}, \mathcal{Y})$ of homomorphisms of ind-objects is $\mathcal{U}$ -small; equivalently, the categories (8.2.4.4) are $\mathcal{U}$ -categories. Equivalently again, by (8.2.4.6), the category $\mathrm{Ind}(C)$ is a $\mathcal{U}$ -category; that is, the right hand side of (8.2.5.1) is small when $I, J \in \mathcal{U}$, which follows immediately from the fact that C is a $\mathcal{U}$ -category, hence the $\mathrm{Hom}_C(X_i, Y_j)$ are small.

8.2.6

Let I be an essentially small filtered category. Then one sees either from (8.2.5.1), or from the functoriality of the inductive limit of a functor $I \rightarrow \widehat{C}$ with respect to that functor, that there is a canonical functor

$$\mathrm{Hom}(I, C) \longrightarrow \mathrm{Ind}(C). \quad (8.2.6.1)$$

Notice that this functor is *not* in general fully faithful, or even faithful.

Remark 8.2.7. The definitions of 8.2.4 make clear the notion of *isomorphism* of two essentially small ind-objects, which also means the isomorphism of the presheaves they ind-represent. Notice that this is a very weak notion of isomorphism when compared with the notion of isomorphism in categories of the form $\mathrm{Hom}(I, C)$. Thus many fairly natural relations one can impose on an ind-object, such as having its components in a given strictly full subcategory, or being strict (8.12 below), are *not* stable under isomorphism. Therefore, if one wants to work with notions stable under isomorphism of ind-objects, one must “saturate” the notions under consideration by passing to the strictly full subcategory of $\mathrm{Ind}(C)$ generated by the subcategory of $\mathrm{Ind}(C)$ consisting of objects satisfying the condition in question. See, for example, the notion of essentially constant ind-object introduced below (8.4).

Exercise 8.2.8. Let $\mathcal{U} \subset \mathcal{V}$ be two universes and let C be a $\mathcal{U}$ -category belonging to $\mathcal{V}$. Let $\mathrm{SysInd}(C)$ denote the following category.

- a) The objects of $\mathrm{SysInd}(C)$ are functors $\varphi : I \rightarrow C$, where I is an essentially $\mathcal{U}$ -small filtered category belonging to $\mathcal{V}$.
- b) If (I, φ) and (J, ψ) are two objects of $\mathrm{SysInd}(C)$, a morphism in $\mathrm{SysInd}(C)$ from (I, φ) to (J, ψ) is a pair (m, u) , where $m : I \rightarrow J$ is a functor and $u : \varphi \rightarrow \psi \circ m$ is a morphism of functors. The composition of morphisms in $\mathrm{SysInd}(C)$ is defined in the evident way.

Let S be the set of morphisms $(m, u) : (I, \varphi) \rightarrow (J, \Psi)$ of $\text{SysInd}(C)$ such that m is cofinal and u is an isomorphism. Let (I, φ) be an object of $\text{SysInd}(C)$. The category $\text{Fl}(I)$ of arrows of I maps to I by two natural functors: source and target. Moreover these two functors are related by the canonical morphism of functors $v : \text{source} \rightarrow \text{target}$. Hence there are two morphisms in $\text{SysInd}(C)$ from $(\text{Fl}(I), \varphi \circ \text{source})$ to (I, φ) :

$$p_1(I, \varphi) = (\text{source}, \text{id}),$$

and

$$p_2(I, \varphi) = (\text{target}, \varphi^* v : \varphi \circ \text{source} \rightarrow \varphi \circ \text{target}).$$

Let $p : \text{SysInd}(C) \rightarrow \text{Ind}(C)$ be the evident functor. Let B be a category. Show that the functor $F \mapsto F \circ p$,

$$\text{Hom}(\text{Ind}(C), B) \longrightarrow \text{Hom}(\text{SysInd}(C), B),$$

is fully faithful, and that a functor $G : \text{SysInd}(C) \rightarrow B$ belongs to its essential image if and only if it has the following two properties:

- 1) For every $s \in S$, $F(s)$ is an isomorphism of B .
- 2) For every object (I, φ) of $\text{SysInd}(C)$, $F(p_1(I, \varphi)) = F(p_2(I, \varphi))$.

8.3 Characterization of ind-representable functors

Proposition 8.3.1. Let F be an ind-representable presheaf on C . Then F is left exact, i.e. (2.3.2), for every finite category J and every functor $\varphi : J \rightarrow C$ such that $\varinjlim \varphi$ is representable (i.e. $\varprojlim \varphi^{\text{op}}$ is representable), the canonical map

$$F(\varinjlim \varphi) \longrightarrow \varprojlim F \circ \varphi$$

is bijective.

Indeed, representable presheaves are plainly left exact; hence by 2.8 so is every filtered inductive limit of such functors. $\square$

8.3.2

Given a presheaf F on C , we shall often have to work with the category

$$C_{/F} \hookrightarrow \widehat{C}_{/F}, \tag{8.3.2.1}$$

which denotes the full subcategory of the second category consisting of arrows $X \rightarrow F$ whose source lies in C . It follows from 1.4 that the objects of this category are identified with pairs (X, u) , where X is an object of C and $u \in F(X)$. A morphism from (X, u) to (X', u') is then interpreted as an arrow $f : X \rightarrow X'$ in C such that $F(f)(u') = u$. The “forget-the-target” functor from $\widehat{C}_{/F}$ to $\widehat{C}$ induces a functor, called the *canonical* functor,

$$C_{/F} \longrightarrow C, \tag{8.3.2.2}$$

already considered in 3.4, where it is proved that the inductive limit of this functor exists in $\widehat{C}$ (with no smallness condition on $C_{/F}$) and is canonically isomorphic to F .

8.3.2.3

Notice that if finite inductive limits are representable in C , and if F is left exact, i.e. carries them to finite projective limits in **Set**, then the same is true in $C_{/F}$, and *a fortiori* (2.7.1) $C_{/F}$ is filtered. More generally, if sums of two objects and cokernels of double arrows are representable in C , and if F carries them respectively to products and to kernels, then $C_{/F}$ is stable under the same types of finite inductive limits. Hence it is filtered if and only if it is nonempty, i.e. if and only if the functor F is not the constant functor with value $\emptyset$.

Theorem 8.3.3. Let C be a $\mathcal{U}$ -category and let $F \in \text{Ob } \widehat{C}$, say $F : C^{\text{op}} \rightarrow (\mathcal{U}\text{-Set})$ is a $\mathcal{U}$ -presheaf on C . The following conditions are equivalent:

- (i) F is ind-representable (8.2.2).
- (ii) The category $C_{/F}$ (8.3.2) is filtered and essentially small.
- (iii) If finite inductive limits are representable in C : the functor F is left exact, and $\text{Ob } C_{/F}$ admits a small cofinal subset (8.1.4).
- (iii bis) If sums of two objects and cokernels of double arrows are representable in C : the functor F carries sums of two objects of C to products and cokernels of double arrows of C to kernels, the category $C_{/F}$ is nonempty, i.e. the functor F is not the constant functor with value $\emptyset$, and finally there exists a small family of objects of $C_{/F}$ such that every object X of $C_{/F}$ is dominated by some X_i , i.e. admits an F -morphism $X_i \rightarrow X$.
- (iv) If the category C is equivalent to a small category: the category $C_{/F}$ is filtered.
- (v) If the category C is equivalent to a small category and if filtered inductive limits in C are representable: the functor F is left exact.

Proof. (i) $\Rightarrow$ (ii). Suppose that F is ind-represented by $(X_i)_{i \in I}$, with I small. We prove that $C_{/F}$ is filtered. Let X, X' be two objects of $C_{/F}$, i.e. objects of C equipped with morphisms $X \rightarrow F$ and $X' \rightarrow F$. We prove that they are dominated by a third object of $C_{/F}$. The morphisms $X \rightarrow F$ and $X' \rightarrow F$ come from morphisms $X \rightarrow X_i$ and $X' \rightarrow X_{i'}$, and after replacing i, i' by a common upper bound in $\text{Ob } I$, we may suppose $i = i'$. We then take as common upper bound of X and X' the object X_i equipped with the canonical morphism $X_i \rightarrow F$.

Now let

$$X \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} X' \xrightarrow{h} F$$

be a double arrow in $C_{/F}$. We prove that it is equalized by an arrow $X' \rightarrow X''$ of $C_{/F}$. The morphism $h : X' \rightarrow F$ is given by a morphism $h_i : X' \rightarrow X_i$, and the condition $hf = hg$ says that there exists $\alpha : i \rightarrow j$ in I such that

$$\varphi(\alpha)(h_i f) = \varphi(\alpha)(h_i g).$$

Thus, after replacing h_i by $\varphi(\alpha)h_i$, we equalize f and g . This proves that $C_{/F}$ is filtered. It is then immediate that it is essentially small, since the objects $(X_i \rightarrow F)_{i \in \text{Ob } I}$ form a small cofinal family in $\text{Ob } C_{/F}$.

(ii) $\Rightarrow$ (i). Put $I = C_{/F}$ and consider the canonical functor (8.3.2.1)

$$\varphi : I = C_{/F} \longrightarrow C.$$

The presheaf represented by this ind-object of C is known to be F (3.4), so F is ind-representable by definition (8.2.2).

(ii) $\Leftrightarrow$ (iii). Indeed, (ii) $\Rightarrow$ (iii) because an ind-representable functor is left exact (8.3.1), and (iii) $\Rightarrow$ (ii) because we have observed (8.3.2.3) that left exactness of F implies that C/F is filtered, and one applies Definition 8.1.8 to conclude that this category is essentially small.

The equivalence (ii) $\Leftrightarrow$ (iii bis) is proved in the same way as (ii) $\Leftrightarrow$ (iii).

The equivalences (ii) $\Leftrightarrow$ (iv) and (iii) $\Leftrightarrow$ (v) are trivial, since if C is equivalent to a small category, then C/F is evidently also equivalent to a small category and *a fortiori* is automatically essentially small as soon as it is filtered. This completes the proof of 8.3.3.

Remark 8.3.4. Let $F : C' \rightarrow C''$ be a functor between $\mathcal{U}$ -categories. It appears that the notion of left exactness (2.3.2) is of practical interest only when finite projective limits are representable in C' . In the case where $C'' = (\mathcal{U}\text{-}\mathbf{Set})$ and C' is $\mathcal{U}$ -small, writing C' in the form C^{op} (so $C = C'^{\text{op}}$), the “right notion” which in the general case replaces left exactness is that of ind-representability of F when it is considered as a presheaf on C^{op} ; equivalently, it is the pro-representability of F when it is considered as a functor on C' (cf. 8.10). This notion does coincide with left exactness when finite projective limits are representable in C' , i.e. when finite inductive limits are representable in C (8.3.3). When one no longer assumes that C' is $\mathcal{U}$ -small, or at least equivalent to a $\mathcal{U}$ -small category, the two notions for a functor F , left exactness and pro-representability, no longer necessarily coincide, even if finite projective limits are representable in C' (cf. 8.12.9). In fact, it seems that in this case left exact functors which are not ind-representable should be regarded as pathological; the “good” objects remain the ind-representable functors. Compare 8.13.3. For the case of functors $F : C' \rightarrow C''$, with C' and C'' again arbitrary $\mathcal{U}$ -categories, we shall develop below (8.11.5), more generally, a notion which “improves” that of left exactness, namely the notion of a functor admitting a pro-adjoint functor.

8.4 Constant and essentially constant ind-objects

Choose a final category e , i.e. one such that $\text{Ob } e$ and $\text{Fl } e$ are each reduced to a single element; for example, one may take for e the full subcategory of $\mathbf{Set}$ consisting of the empty set, if one wants a canonical choice. This is plainly a filtered and small category. If X is an object of C , we associate with it the constant ind-object indexed by e , with value X , denoted $c(X)$. It is clear that, as X varies, one obtains a fully faithful functor

$$c : C \longrightarrow \text{Ind}(C), \quad (8.4.1)$$

which is moreover injective on objects, and through which we shall identify C with a full subcategory of $\text{Ind}(C)$. Notice that the composite functor

$$C \xrightarrow{c} \text{Ind}(C) \xrightarrow{L} \widehat{C} \quad (8.4.2)$$

is the canonical functor h (8.2.1.1).

8.4.3

More generally, let I be a filtered category. The constant functor on I with value an object X of C is also called *the constant ind-object of value X indexed by I* . If I is essentially small, it is clear that this ind-object ind-represents the functor $h(X)$ represented by X , hence this ind-object is isomorphic to $c(X)$.

8.4.4

An ind-object $\mathcal{X}$ of C is called *essentially constant* if it is isomorphic (in a category $\text{Ind}_{\mathcal{V}}(C, \mathcal{V}')$ for suitable $\mathcal{V}, \mathcal{V}'$) to an ind-object of the form $c(X)$, with $X \in \text{Ob } C$. The object X , which is then determined up to canonical isomorphism, is called the *value* of the essentially constant ind-object in question. Thus, by definition, the functor c of (8.4.1) induces an equivalence of C with the full subcategory of $\text{Ind}(C)$ consisting of essentially constant ind-objects; this subcategory is the essential image of the functor c .

Obviously a constant ind-object is essentially constant; the converse is true only in the trivial case where C is empty or a one-point category.

8.5 Filtered inductive limits in $\text{Ind}(C)$

Proposition 8.5.1. Let C be a $\mathcal{U}$ -category. In $\text{Ind}(C)$, small filtered inductive limits are representable, and the canonical functor (8.2.4.8)

$$L : \text{Ind}(C) \longrightarrow \widehat{C}$$

commutes with them.

Taking account of the fact that L is fully faithful, the assertions of the proposition are equivalent to the following one.

Corollary 8.5.2. Every small filtered inductive limit, in $\widehat{C}$, of ind-representable presheaves is ind-representable.

This follows easily from criterion 8.3.3(ii). The details of the verification are left to the reader.

8.5.3

The “tautological” case of filtered inductive limits in $\text{Ind}(C)$ is the case where one starts from an essentially small ind-object $\mathcal{X} = (X_i)_{i \in I}$ of C . One then has, in $\widehat{C}$ and hence also in $\text{Ind}(C)$,

$$\mathcal{X} = \varinjlim_I X_i. \tag{8.5.3.1}$$

In writing this formula, one should take care that this is not an inductive limit in C , and that even when the latter exists, it is not isomorphic in $\text{Ind}(C)$ to $\mathcal{X}$: indeed, the canonical functor (8.4.1) $c : C \rightarrow \text{Ind}(C)$ *does not in general commute with filtered inductive limits*. See Exercise 8.5.5 below.

To avoid this possible confusion in writing (8.5.3.1), “certain authors” (= P. Deligne) prefer to write it

$$\mathcal{X} = “\varinjlim_I” X_i, \tag{8.5.3.2}$$

the role of the quotation marks being to indicate that the inductive limit is taken in a category of ind-objects $\text{Ind}(C)$. By extension, one should then denote by “lim” every inductive-limit operation in $\text{Ind}(C)$, without requiring the components of the inductive system considered in $\text{Ind}(C)$ to lie in the image, or in the essential image, of $c : C \rightarrow \text{Ind}(C)$.

8.5.4

To compute the inductive or projective limit of a functor

$$\varphi : J \longrightarrow \text{Ind}(C), \quad (8.5.4.1)$$

where J is now an arbitrary category, it is often convenient to make φ explicit by means of a functor

$$\Phi : J \times I \longrightarrow C, \quad (8.5.4.2)$$

where I is an essentially small, or preferably small, filtered category. Such a Φ indeed defines a functor

$$j \longmapsto (i \longmapsto \Phi(j, i)) : J \longrightarrow \text{Hom}(I, C),$$

hence, after composing with the canonical arrow (8.2.6.1) $\text{Hom}(I, C) \rightarrow \text{Ind}(C)$, a functor

$$j \longmapsto (i \longmapsto \Phi(j, i)) : J \longrightarrow \text{Ind}(C). \quad (8.5.4.3)$$

We may say that Φ is an *indexed expression of φ , indexed by the filtered category I* , if the corresponding functor (8.5.4.3) is isomorphic to φ . We shall study below (8.8) general existence conditions for an indexed expression of a given functor φ . Here we shall merely start from a functor φ given in indexed form (8.5.4.3), in the case where J is a small filtered category, in order to indicate the calculation of

$$\varinjlim \varphi = \varinjlim_j \varphi(j).$$

Formula (8.5.3.2), together with the associativity formula for inductive limits (2.5.0), then immediately gives the *tautological calculation* of $\varinjlim \varphi$:

$$\varinjlim_J \varphi = \varinjlim_{J \times I} \Phi, \quad (8.5.4.4)$$

i.e.

$$\varinjlim_j (i \longmapsto \Phi(j, i)) = \varinjlim_{j, i} \Phi(j, i). \quad (8.5.4.5)$$

Thus the inductive system sought is none other than Φ itself, the index category being $J \times I$, which is indeed filtered since J and I are so.

Exercise 8.5.5. Let C be a $\mathcal{U}$ -category.

a) Prove that the following conditions are equivalent:

- (i) Small inductive limits are representable in C , and the functor $c : C \rightarrow \text{Ind}(C)$ commutes with them.
- (ii) Small inductive limits are representable in C , and for every $X \in \text{Ob } C$, the functor $Y \mapsto \text{Hom}(X, Y)$ commutes with these limits.
- (iii) The functor c is an equivalence of categories.
- (iv) Small filtered inductive limits are representable in C , and the functor $\lim c : \text{Ind } C \rightarrow C$ (cf. 8.7.1.5) is fully faithful, or equivalently an equivalence.

b) If C is a finite category, prove that the preceding conditions are equivalent to the following one: for every projector in C (10.6), its image is representable in C .

8.6 Extension of a functor to ind-objects

8.6.1

Let

$$f : C \longrightarrow C' \quad (8.6.1.1)$$

be a functor between $\mathcal{U}$ -categories. For every ind-object

$$\varphi : I \longrightarrow C$$

of C , where I is a filtered category, the composite functor $f \circ \varphi$ is an ind-object of C' , also denoted $\text{Ind}(f)(\varphi)$. In indexed notation, if φ is written as $\mathcal{X} = (X_i)_{i \in I}$, one obtains

$$\text{Ind}(f)(X_i)_{i \in I} = (f(X_i))_{i \in I}. \quad (8.6.1.2)$$

Let $\mathcal{Y} = (Y_j)_{j \in J}$ be a second inductive system of C . Then one obtains an evident map, also denoted $u \mapsto \text{Ind}(f)(u)$:

$$\begin{aligned} \text{Hom}(\mathcal{X}, \mathcal{Y}) &= \varprojlim_i \varinjlim_j \text{Hom}(X_i, Y_j) \\ &\longrightarrow \text{Hom}(\text{Ind}(f)(\mathcal{X}), \text{Ind}(f)(\mathcal{Y})) \simeq \varprojlim_i \varinjlim_j \text{Hom}(f(X_i), f(Y_j)). \end{aligned} \quad (8.6.1.3)$$

One immediately checks that these maps are compatible with the composition of morphisms of ind-objects, referring to the unwritten formula for composition of morphisms of ind-objects (8.2.5). We have thus obtained, for every universe $\mathcal{V}$, a functor

$$\text{Ind}(f) : \text{Ind}_{\mathcal{V}}(C, \mathcal{U}) \longrightarrow \text{Ind}_{\mathcal{V}}(C', \mathcal{U}) \quad (8.6.1.4)$$

(notation of (8.2.4.5)), and in particular a functor

$$\text{Ind}(f) : \text{Ind}(C) \longrightarrow \text{Ind}(C'). \quad (8.6.1.5)$$

If one has a second functor

$$g : C' \longrightarrow C'',$$

then of course there is an identity of functors between categories of ind-objects, or of $\text{Ind}_{\mathcal{V}}$ -objects if desired,

$$\text{Ind}(gf) = \text{Ind}(g) \circ \text{Ind}(f), \quad (8.6.1.6)$$

and for $C' = C$ one has the perennial formula

$$\text{Ind}(\text{id}_C) = \text{id}_{\text{Ind}(C)}. \quad (8.6.1.7)$$

Thus, figuratively speaking, one may say that the category $\text{Ind}(C)$ *depends functorially on C* , covariantly.¹ We leave to the reader the task of making explicit even a 2-*functorial* dependence, by defining, for every pair of $\mathcal{U}$ -categories C, C' , a canonical functor

$$\text{Ind} : \text{Hom}(C, C') \longrightarrow \text{Hom}(\text{Ind}(C), \text{Ind}(C')), \quad (8.6.1.8)$$

and by specifying the functorial nature of the identity isomorphisms (8.6.1.6) and (8.6.1.7).

¹For a contravariant dependence, under certain conditions, cf. 8.11.

8.6.2

Return to the functor (8.6.1.1), and consider the corresponding functor (5.1)

$$f_! : \widehat{C} \longrightarrow \widehat{C'}, \quad (8.6.2.1)$$

and the diagram of functors

$$\begin{array}{ccc} \text{Ind}(C) & \xrightarrow{\text{Ind}(f)} & \text{Ind}(C') \\ L_C \downarrow & & \downarrow L_{C'} \\ \widehat{C} & \xrightarrow{f_!} & \widehat{C'}. \end{array} \quad (8.6.2.2)$$

Here the vertical arrows denote the canonical functors L of (8.2.4.8). Since the functor $f_!$ “extends f ” and commutes with inductive limits (5.4.3), and since the functors L_C and $L_{C'}$ commute with small filtered inductive limits (8.5.1), it follows from (8.5.3.2) that for every ind-object $(X_i)_{i \in I}$ of C there is, in $\widehat{C'}$, a canonical isomorphism

$$f_!(L_C((X_i)_{i \in I})) \simeq \varinjlim_i f_! L_C(X_i) \simeq \varinjlim_i L_{C'} f(X_i) = L_{C'}(f(X_i)_{i \in I}),$$

i.e. a canonical isomorphism

$$f_! L_C(\mathcal{X}) \simeq L_{C'}(\text{Ind}(f)(\mathcal{X})).$$

We leave to the reader the verification that the latter is functorial, i.e. that it corresponds to a canonical isomorphism

$$f_! L_C \simeq L_{C'} \circ \text{Ind}(f). \quad (8.6.2.3)$$

In other words, *the square (8.6.2.2) is commutative up to canonical isomorphism*. Taking account of the fact that $f_!$ commutes with small inductive limits, and of 8.5.1, we conclude as follows.

Proposition 8.6.3. Let $f : C \rightarrow C'$ be a functor between $\mathcal{U}$ -categories. Then the functor

$$\text{Ind}(f) : \text{Ind}(C) \longrightarrow \text{Ind}(C')$$

commutes with filtered inductive limits. Moreover, it makes the following diagram commutative:

$$\begin{array}{ccc} C & \xrightarrow{f} & C' \\ c_C \downarrow & & \downarrow c_{C'} \\ \text{Ind}(C) & \xrightarrow{\text{Ind}(f)} & \text{Ind}(C'), \end{array}$$

where the vertical arrows $c_C, c_{C'}$ are the canonical functors (8.4.1).

The last assertion is trivial from the definitions and has been included for ease of reference. Note moreover that the properties stated in 8.6.3 *characterize the functor* $\text{Ind}(f)$ up to unique isomorphism as being induced by the functor $f_!$ (8.6.2.1), as follows from the proof just given of (8.6.2.3).

Proposition 8.6.4. The notation is that of 8.6.3.

- a) For $\text{Ind}(f)$ to be faithful, respectively fully faithful, it is necessary and sufficient that f be so.

- b) For $\text{Ind}(f)$ to be an equivalence of categories, it is necessary and sufficient that f be fully faithful and that every object of C' be isomorphic to a direct factor (10.6) of an object in the image of f .

Proof.

- a) Necessity follows immediately from the fact that f is induced by $\text{Ind}(f)$. For sufficiency, it is enough to use the form (8.6.1.3) of $\text{Ind}(f)$ on the sets $\text{Hom}(\mathcal{X}, \mathcal{Y})$, remembering that filtered inductive limits of sets, and arbitrary projective limits, carry monomorphisms to monomorphisms and isomorphisms to isomorphisms.
- b) We may already suppose f , hence $\text{Ind}(f)$, fully faithful. Since every object of $\text{Ind}(C')$ is a small filtered inductive limit of objects of C' , full faithfulness of $\text{Ind}(f)$ implies that for this functor to be essentially surjective it is equivalent that every object X' of C' lie in its essential image. If one has an isomorphism

$$X' \xrightarrow{\sim} \varinjlim f(X_i),$$

then this isomorphism factors through one of the $f(X_i)$; this shows that X' is a direct factor of $f(X_i)$, proving necessity in b). For sufficiency, using the full faithfulness of $\text{Ind}(f)$, the assertion follows at once from the following corollary.

Corollary 8.6.5. In $\text{Ind}(C)$ the images of projectors (10.6) are representable, and the functor $\text{Ind}(f) : \text{Ind}(C) \rightarrow \text{Ind}(C')$ commutes with the formation of these images.

Indeed, this follows from the fact that in $\text{Ind}(C)$ small filtered inductive limits are representable (8.5.1) and that $\text{Ind}(f)$ commutes with them (8.6.3), taking account of the fact that the image of a projector $p : X \rightarrow X$ may be interpreted as the inductive limit of the filtered inductive system indexed by $\mathbb{N}$

$$X \xrightarrow{p} X \xrightarrow{p} X \longrightarrow \cdots,$$

or, equivalently, as the inductive limit of the evident functor on the filtered category P having a single object and a nonidentity arrow Π such that $\Pi^2 = \Pi$.